Active Calculus

Activities Workbook chs 5-8, 2nd Edition

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Matthew Boelkins

David Austin Christina Safranski Steven Schlicker

Activities Workbook
for
Active Calculus Single Variable
Chapters 5–8, 2nd Edition

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Matthew Boelkins
Grand Valley State University

Contributing Authors

David Austin
Grand Valley State University

Christina Safranski
Franciscan University of Steubenville

Steven Schlicker
Grand Valley State University

Production Editor

Mitchel T. Keller
University of Wisconsin-Madison

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Preface

This Activities Workbook for *Active Calculus Single Variable* collects all the Preview Activities and Activities in a way that each starts on a new page. The Activities Workbook provides space for students to write and organize their work as they complete the activities found in the text.

The design of this workbook is such that each Preview Activity and Activity starts on a right-hand page. The workbook is offered in two volumes: the first for chapters 1–4 and the second for chapters 5–8. This workbook is for chapters 5–8.

Contents

Preface v

5 Evaluating Integrals 1

5.1 Constructing accurate graphs of antiderivatives . . . 1
5.2 The Second Fundamental Theorem of Calculus . . . 9
5.3 Integration by substitution . . . 17
5.4 Integration by parts . . . 25
5.5 Other options for finding algebraic antiderivatives . . . 33
5.6 Numerical integration . . . 39

6 Using Definite Integrals 47

6.1 Using definite integrals to find area and length . . . 47
6.2 Using definite integrals to find volume . . . 55
6.3 Density, mass, and center of mass . . . 67
6.4 Physics applications: work, force, and pressure . . . 75
6.5 Improper integrals . . . 83

7 Differential Equations 91

7.1 An introduction to differential equations . . . 91
7.2 Qualitative behavior of solutions to differential equations . . . 99
7.3 Euler's method. . . . 105
7.4 Separable differential equations . . . 111
7.5 Modeling with differential equations. . . . 119
7.6 Population growth and the logistic equation . . . 125

8 Taylor Polynomials and Taylor Series 131

8.1 Extending local linearization. . . . 131
8.2 Taylor polynomials . . . 139
8.3 Geometric sums . . . 147
8.4 Taylor series. . . . 155

8.5 Finding and using Taylor series 161
8.6 Quantifying the accuracy of approximations 171

Appendices

A A Short Table of Integrals **177**

Back Matter

CHAPTER 5

Evaluating Integrals

5.1 Constructing accurate graphs of antiderivatives

Preview Activity 5.1.1. Suppose that the following information is known about a function f: the graph of its derivative, $y = f'(x)$, is given in Figure 5.1.1. Further, assume that f' is piecewise linear (as pictured) and that for $x \leq 0$ and $x \geq 6$, $f'(x) = 0$. Finally, it is given that $f(0) = 1$.

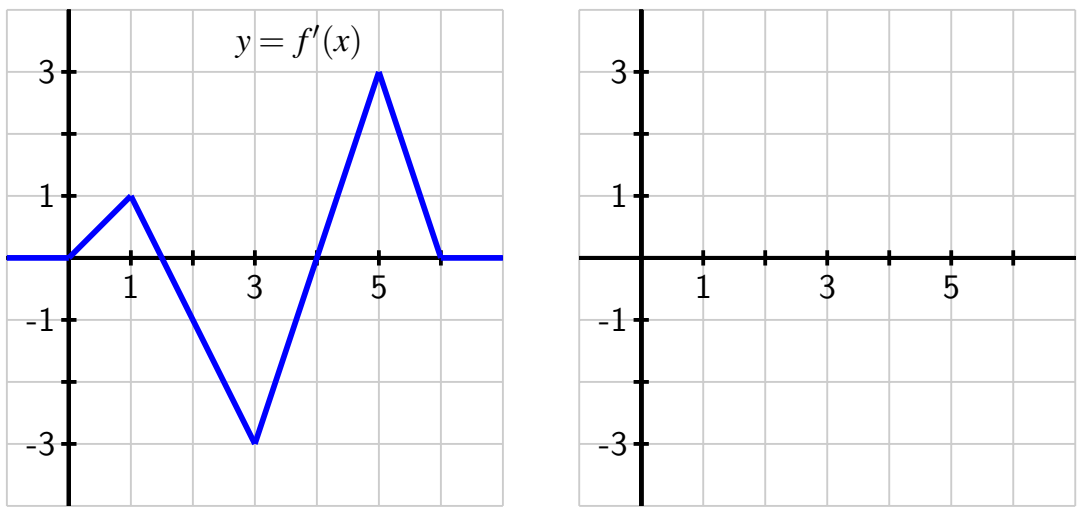


Figure 5.1.1 At left, the graph of $y = f'(x)$; at right, axes for plotting $y = f(x)$.

(a) On what interval(s) is f an increasing function? On what intervals is f decreasing?

(b) On what interval(s) is f concave up? concave down?

(c) At what point(s) does f have a relative minimum? a relative maximum?

(d) Recall that the Total Change Theorem tells us that

$$f(1) - f(0) = \int_0^1 f'(x)\,dx.$$

What is the exact value of $f(1)$?

(e) Use the given information and similar reasoning to that in (d) to determine the exact value of $f(2)$, $f(3)$, $f(4)$, $f(5)$, and $f(6)$.

(f) Based on your responses to all of the preceding questions, sketch a complete and accurate graph of $y = f(x)$ on the axes provided, being sure to indicate the behavior of f for $x < 0$ and $x > 6$.

Activity 5.1.2. Suppose that the function $y = f(x)$ is given by the graph shown in Figure 5.1.2, and that the pieces of f are either portions of lines or portions of circles. In addition, let F be an antiderivative of f and say that $F(0) = -1$. Finally, assume that for $x \leq 0$ and $x \geq 7$, $f(x) = 0$.

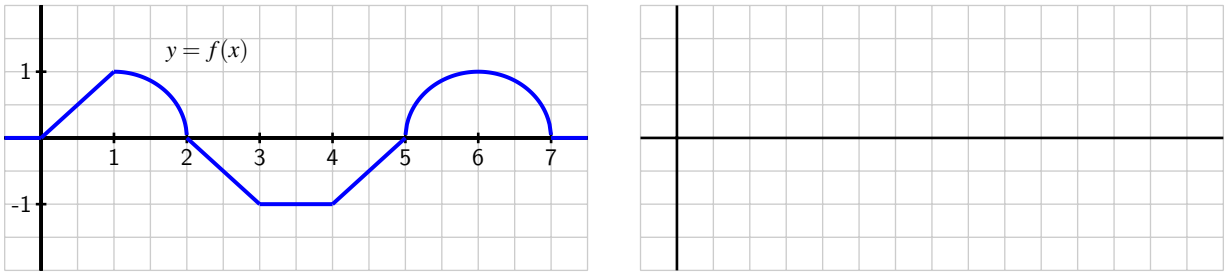


Figure 5.1.2 At left, the graph of $y = f(x)$.

(a) On what interval(s) is F an increasing function? On what intervals is F decreasing?

(b) On what interval(s) is F concave up? concave down? neither?

(c) At what point(s) does F have a relative minimum? a relative maximum?

(d) Use the given information to determine the exact value of $F(x)$ for $x = 1, 2, \ldots, 7$. In addition, what are the values of $F(-1)$ and $F(8)$?

(e) Based on your responses to all of the preceding questions, sketch a complete and accurate graph of $y = F(x)$ on the axes provided, being sure to indicate the behavior of F for $x < 0$ and $x > 7$. Clearly indicate the scale on the vertical and horizontal axes of your graph.

(f) Suppose we change one key piece of information: in particular, say that G is an antiderivative of f and $G(0) = 0$. How (if at all) would your answers to the preceding questions change? Sketch a graph of G on the same axes as the graph of F you constructed in (e).

Activity 5.1.3. For each of the following functions, sketch an accurate graph of the antiderivative that satisfies the given initial condition. In addition, sketch the graph of two additional antiderivatives of the given function, and state the corresponding initial conditions that each of them satisfy. If possible, find an algebraic formula for the antiderivative that satisfies the initial condition.

(a) original function: $g(x) = |x| - 1$; initial condition: $G(-1) = 0$; interval for sketch: $[-2, 2]$

(b) original function: $h(x) = \sin(x)$; initial condition: $H(0) = 1$; interval for sketch: $[0, 4\pi]$

(c) original function: $p(x) = \begin{cases} x^2, & \text{if } 0 < x < 1 \\ -(x-2)^2, & \text{if } 1 < x < 2 \\ 0 & \text{otherwise} \end{cases}$; initial condition: $P(0) = 1$; interval for sketch: $[-1, 3]$

Activity 5.1.4. Suppose that g is given by the graph at left in Figure 5.1.3 and that A is the corresponding integral function defined by $A(x) = \int_1^x g(t)\,dt$.

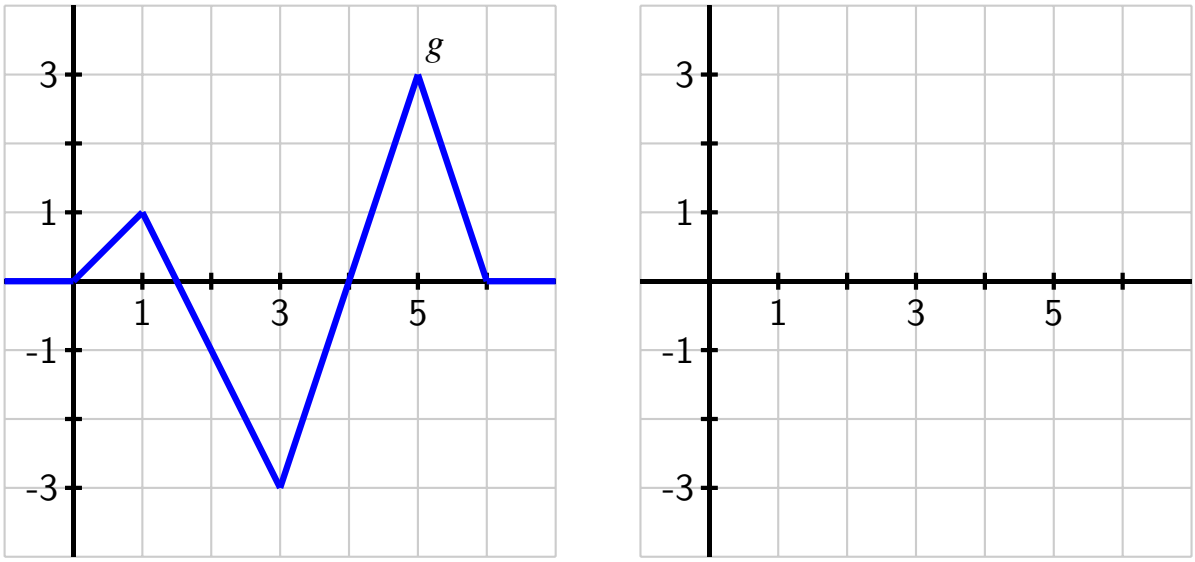


Figure 5.1.3 At left, the graph of $y = g(t)$; at right, axes for plotting $y = A(x)$, where A is defined by the formula $A(x) = \int_1^x g(t)\,dt$.

(a) On what interval(s) is A an increasing function? On what intervals is A decreasing? Why?

(b) On what interval(s) do you think A is concave up? concave down? Why?

(c) At what point(s) does A have a relative minimum? a relative maximum?

(d) Use the given information to determine the exact values of $A(0)$, $A(1)$, $A(2)$, $A(3)$, $A(4)$, $A(5)$, and $A(6)$.

(e) Based on your responses to all of the preceding questions, sketch a complete and accurate graph of $y = A(x)$ on the axes provided, being sure to indicate the behavior of A for $x < 0$ and $x > 6$.

(f) How does the graph of B compare to A if B is instead defined by $B(x) = \int_0^x g(t)\,dt$?

5.2 The Second Fundamental Theorem of Calculus

Preview Activity 5.2.1. Consider the function A defined by the rule

$$A(x) = \int_1^x f(t)\,dt,$$

where $f(t) = 4 - 2t$.

(a) Compute $A(1)$ and $A(2)$ exactly.

(b) Use the First Fundamental Theorem of Calculus to find a formula for $A(x)$ that does not involve an integral. That is, use the first FTC to evaluate $\int_1^x (4 - 2t)\,dt$.

(c) Observe that f is a linear function; what kind of function is A?

(d) Using the formula you found in (b) that does not involve an integral, compute $A'(x)$.

(e) While we have defined f by the rule $f(t) = 4 - 2t$, it is equivalent to say that f is given by the rule $f(x) = 4 - 2x$. What do you observe about the relationship between A and f?

Activity 5.2.2. Suppose that f is the function given in Figure 5.2.1 and that f is a piecewise function whose parts are either portions of lines or portions of circles, as pictured.

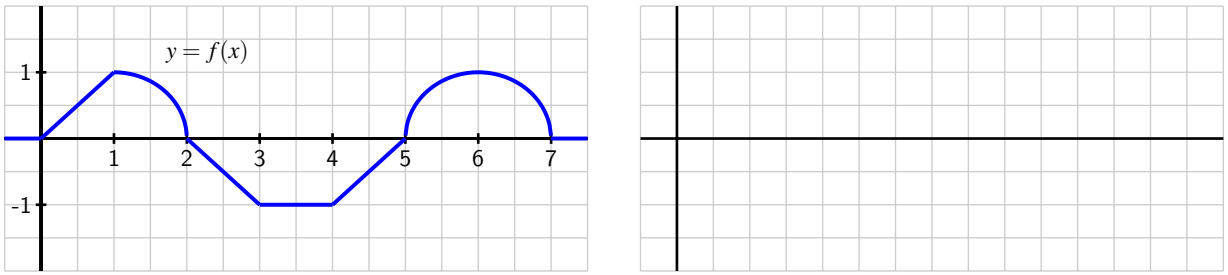


Figure 5.2.1 At left, the graph of $y = f(x)$. At right, axes for sketching $y = A(x)$.

In addition, let A be the function defined by the rule $A(x) = \int_2^x f(t)\,dt$.

(a) What does the Second FTC tell us about the relationship between A and f?

(b) Compute $A(1)$ and $A(3)$ exactly.

(c) Sketch a precise graph of $y = A(x)$ on the axes at right that accurately reflects where A is increasing and decreasing, where A is concave up and concave down, and the exact values of A at $x = 0, 1, \ldots, 7$.

(d) How is A similar to, but different from, the function F that you found in Activity 5.1.2?

(e) With as little additional work as possible, sketch precise graphs of the functions $B(x) = \int_3^x f(t)\,dt$ and $C(x) = \int_1^x f(t)\,dt$. Justify your results with at least one sentence of explanation.

Activity 5.2.3. Suppose that $f(t) = \frac{t}{1+t^2}$.

Note that while we know a function whose derivative is $\frac{1}{1+t^2}$, namely $\arctan(t)$, we haven't yet encountered an elementary function whose derivative is $f(t) = \frac{t}{1+t^2}$.

Let $F(x) = \int_0^x f(t)\,dt$. We will explore $F(x)$ from numerical, graphical, and algebraic perspectives.

(a) What is the key relationship between F and f, according to the Second FTC?

(b) Analyze the first derivative of F algebraically to determine the intervals on which F is increasing and decreasing.

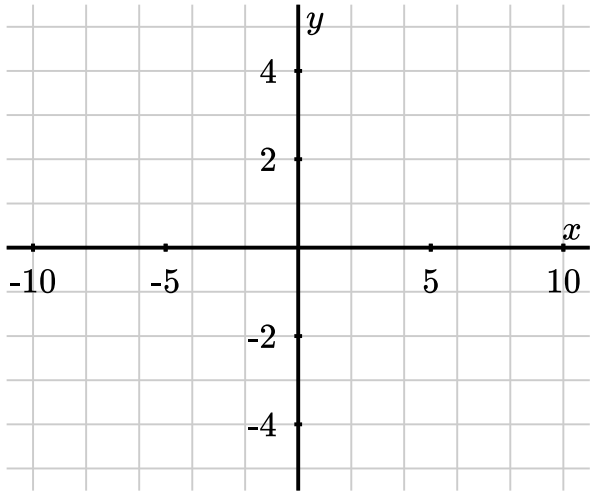
(c) Analyze the second derivative of F algebraically to determine the intervals on which F is concave up and concave down. Note that $f'(t)$ can be simplified to be written in the form $f'(t) = \frac{1-t^2}{(1+t^2)^2}$.

(d) Use a Riemann sum calculator with midpoints and 10 subintervals to calculate approximate values of $F(x)$ in the table shown below.

Note that $F(0) = \int_0^0 f(t)\,dt = 0$.

x	-10	-5	0	5	10
$F(x)$	_____	_____	0	_____	_____

(e) Use your work above to sketch an accurate graph of $y = F(x)$ on the axes provided.



(f) Now consider the function $g(x) = \frac{1}{2}\ln(1 + x^2)$. Calculate $g(0)$ and $g'(x)$. Use appropriate computing technology to plot the graph of $g(x)$. What can you conclude about the functions $F(x)$ and $g(x)$?

Activity 5.2.4. Evaluate each of the following derivatives and definite integrals. Clearly cite whether you use the First or Second FTC in so doing.

(a) $\frac{d}{dx}\left[\int_4^x e^{t^2}\,dt\right]$

(b) $\int_{-2}^x \frac{d}{dt}\left[\frac{t^4}{1+t^4}\right]\,dt$

(c) $\frac{d}{dx}\left[\int_x^1 \cos(t^3)\,dt\right]$

(d) $\int_3^x \frac{d}{dt}\left[\ln(1+t^2)\right]\,dt$

(e) $\frac{d}{dx}\left[\int_4^{x^3} \sin(t^2)\,dt\right]$

5.3 Integration by substitution

Preview Activity 5.3.1. In Section 2.5, we learned the Chain Rule and how it can be applied to find the derivative of a composite function. In particular, if u is a differentiable function of x, and f is a differentiable function of $u(x)$, then

$$\frac{d}{dx}\left[f(u(x))\right] = f'(u(x)) \cdot u'(x).$$

In words, we say that the derivative of a composite function $c(x) = f(u(x))$, where f is considered the "outer" function and u the "inner" function, is "the derivative of the outer function, evaluated at the inner function, times the derivative of the inner function."

(a) For each of the following functions, use the Chain Rule to find the function's derivative. Be sure to label each derivative by name (e.g., the derivative of $g(x)$ should be labeled $g'(x)$).

i. $g(x) = e^{3x}$

ii. $h(x) = \sin(5x + 1)$

iii. $p(x) = \arctan(2x)$

iv. $q(x) = (2 - 7x)^4$

v. $r(x) = 3^{4-11x}$

(b) For each of the following functions, use your work in (a) to help you determine the general antiderivative (that is, include "+C") of the function. Label each antiderivative by name (e.g., the antiderivative of m should be called M). In addition, check your work by computing the derivative of each proposed antiderivative.

i. $m(x) = e^{3x}$

ii. $n(x) = \cos(5x + 1)$

iii. $s(x) = \dfrac{1}{1 + 4x^2}$

iv. $v(x) = (2 - 7x)^3$

v. $w(x) = 3^{4-11x}$

(c) Based on your experience in parts (a) and (b), conjecture an antiderivative for each of the following functions. Test your conjectures by computing the derivative of each proposed antiderivative.

i. $a(x) = \cos(\pi x)$

ii. $b(x) = (4x + 7)^{11}$

iii. $c(x) = xe^{x^2}$

Activity 5.3.2. Evaluate each of the following indefinite integrals. Check each antiderivative that you find by differentiating.

(a) $\int \sin(8 - 3x)\, dx$

(b) $\int \sec^2(4x)\, dx$

(c) $\int \frac{1}{11x-9}\, dx$

(d) $\int \csc(2x+1)\cot(2x+1)\,dx$

(e) $\int \frac{1}{\sqrt{1-16x^2}}\,dx$

(f) $\int 5^{-x}\,dx$

Activity 5.3.3. Evaluate each of the following indefinite integrals by using these steps:

- Find two functions within the integrand that form (up to a possible missing constant) a function-derivative pair;
- Make a substitution and convert the integral to one involving u and du;
- Evaluate the new integral in u;
- Convert the resulting function of u back to a function of x by using your earlier substitution;
- Check your work by differentiating the function of x. You should come up with the integrand originally given.

(a) $\int \frac{x^2}{5x^3+1}\,dx$

(b) $\int e^x \sin(e^x)\,dx$

(c) $\int \frac{\cos(\sqrt{x})}{\sqrt{x}}\, dx$

Activity 5.3.4. Evaluate each of the following definite integrals exactly through an appropriate u-substitution.

(a) $\int_1^2 \frac{x}{1+4x^2}\,dx$

(b) $\int_0^1 e^{-x}(2e^{-x}+3)^9\,dx$

(c) $\int_{2/\pi}^{4/\pi} \frac{\cos\left(\frac{1}{x}\right)}{x^2}\, dx$

5.4 Integration by parts

Preview Activity 5.4.1. In Section 2.3, we developed the Product Rule and studied how it is employed to differentiate a product of two functions. In particular, recall that if f and g are differentiable functions of x, then

$$\frac{d}{dx}\left[f(x)\cdot g(x)\right] = f(x)\cdot g'(x) + g(x)\cdot f'(x).$$

(a) For each of the following functions, use the Product Rule to find the function's derivative. Be sure to label each derivative by name (e.g., the derivative of $g(x)$ should be labeled $g'(x)$).

i. $g(x) = x\sin(x)$

ii. $h(x) = xe^x$

iii. $p(x) = x\ln(x)$

iv. $q(x) = x^2\cos(x)$

v. $r(x) = e^x\sin(x)$

(b) Use your work in (a) to help you evaluate the following indefinite integrals. Use differentiation to check your work.

i. $\displaystyle\int xe^x + e^x\,dx$

ii. $\displaystyle\int e^x(\sin(x)+\cos(x))\,dx$

iii. $\displaystyle\int 2x\cos(x) - x^2\sin(x)\,dx$

iv. $\displaystyle\int x\cos(x) + \sin(x)\,dx$

v. $\displaystyle\int 1 + \ln(x)\,dx$

(c) Observe that the examples in (b) work nicely because of the derivatives you were asked to calculate in (a). Each integrand in (b) is precisely the result of differentiating one of the products of basic functions found in (a). To see what happens when an integrand is still a product but not necessarily the result of differentiating an elementary product, we consider how to evaluate

$$\int x\cos(x)\,dx.$$

i. First, observe that

$$\frac{d}{dx}\left[x\sin(x)\right] = x\cos(x) + \sin(x).$$

Integrating both sides indefinitely and using the fact that the integral of a sum is the sum of the integrals, we find that

$$\int \left(\frac{d}{dx}\left[x\sin(x)\right]\right)dx = \int x\cos(x)\,dx + \int \sin(x)\,dx.$$

In this last equation, evaluate the indefinite integral on the left side as well as the rightmost indefinite integral on the right.

ii. In the most recent equation from (i.), solve the equation for the expression $\int x\cos(x)\,dx$.

iii. For which product of basic functions have you now found the antiderivative?

Activity 5.4.2. Evaluate each of the following indefinite integrals. Check each antiderivative that you find by differentiating.

(a) $\int te^{-t}\,dt$

(b) $\int 4x\sin(3x)\,dx$

(c) $\int z\sec^2(z)\,dz$

(d) $\int x \ln(x)\, dx$

Activity 5.4.3. Evaluate each of the following indefinite integrals, using the provided hints.

(a) Evaluate $\int \arctan(x)\,dx$ by using integration by parts with the substitution $u = \arctan(x)$ and $dv = 1\,dx$.

(b) Evaluate $\int \ln(z)\,dz$. Consider a similar substitution to the one in (a).

(c) Use the substitution $z = t^2$ to transform the integral $\int t^3 \sin(t^2)\,dt$ to a new integral in the variable z, and evaluate that new integral by parts.

(d) Evaluate $\int s^5 e^{s^3}\, ds$ using an approach similar to that described in (c).

(e) Evaluate $\int e^{2t} \cos(e^t)\, dt$. You will find it helpful to note that $e^{2t} = e^t \cdot e^t$.

Activity 5.4.4. Evaluate each of the following indefinite integrals.

(a) $\int x^2 \sin(x)\, dx$

(b) $\int t^3 \ln(t)\, dt$

(c) $\int e^z \sin(z)\, dz$

(d) $\int s^2 e^{3s}\, ds$

(e) $\int t \arctan(t)\, dt$ (*Hint:* At a certain point in this problem, it is very helpful to note that $\frac{t^2}{1+t^2} = 1 - \frac{1}{1+t^2}$.)

5.5 Other options for finding algebraic antiderivatives

Preview Activity 5.5.1. For each of the indefinite integrals below, the main question is to decide whether the integral can be evaluated using u-substitution, integration by parts, a combination of the two, or neither. For integrals for which your answer is affirmative, state the substitution(s) you would use. It is not necessary to actually evaluate any of the integrals completely, unless the integral can be evaluated immediately using a familiar basic antiderivative.

(a) $\int x^2 \sin(x^3)\, dx$, $\int x^2 \sin(x)\, dx$, $\int \sin(x^3)\, dx$, $\int x^5 \sin(x^3)\, dx$

(b) $\int \frac{1}{1+x^2}\, dx$, $\int \frac{x}{1+x^2}\, dx$, $\int \frac{2x+3}{1+x^2}\, dx$, $\int \frac{e^x}{1+(e^x)^2}\, dx$,

(c) $\int x\ln(x)\,dx$, $\int \frac{\ln(x)}{x}\,dx$, $\int \ln(1+x^2)\,dx$, $\int x\ln(1+x^2)\,dx$,

(d) $\int x\sqrt{1-x^2}\,dx$, $\int \frac{1}{\sqrt{1-x^2}}\,dx$, $\int \frac{x}{\sqrt{1-x^2}}\,dx$, $\int \frac{1}{x\sqrt{1-x^2}}\,dx$,

Activity 5.5.2. For each of the following problems, evaluate the integral by using the partial fraction decomposition provided.

(a) $\int \frac{1}{x^2-2x-3}\,dx$, given that $\frac{1}{x^2-2x-3} = \frac{1/4}{x-3} - \frac{1/4}{x+1}$

(b) $\int \frac{x^2+1}{x^3-x^2}\,dx$, given that $\frac{x^2+1}{x^3-x^2} = -\frac{1}{x} - \frac{1}{x^2} + \frac{2}{x-1}$

(c) $\int \frac{x-2}{x^4+x^2}\,dx$, given that $\frac{x-2}{x^4+x^2} = \frac{1}{x} - \frac{2}{x^2} + \frac{-x+2}{1+x^2}$

Activity 5.5.3. For each of the following integrals, evaluate the integral using u-substitution and/or an entry from the table found in Appendix A.

(a) $\int \sqrt{x^2+4}\,dx$

(b) $\int \frac{x}{\sqrt{x^2+4}}\,dx$

(c) $\int \frac{2}{\sqrt{16+25x^2}}\,dx$

(d) $\int \frac{1}{x^2\sqrt{49-36x^2}}\,dx$

5.6 Numerical integration

Preview Activity 5.6.1. As we begin to investigate ways to approximate definite integrals, it will be insightful to compare results to integrals whose exact values we know. To that end, the following sequence of questions centers on $\int_0^3 x^2\,dx$.

(a) Use the interactive (`gvsu.edu/s/a9`) with the function $f(x) = x^2$ on the window of x values from 0 to 3 to compute L_3, the left Riemann sum with three subintervals.

(b) Likewise, use the interactive to compute R_3 and M_3, the right and middle Riemann sums with three subintervals, respectively.

(c) Use the Fundamental Theorem of Calculus to compute the exact value of $I = \int_0^3 x^2\,dx$.

(d) We define the *error* that results from an approximation of a definite integral to be the approximation's value minus the integral's exact value. What is the error that results from using L_3? From R_3? From M_3?

(e) In what follows in this section, we will learn a new approach to estimating the value of a definite integral known as the Trapezoid Rule. The basic idea is to use trapezoids, rather than rectangles, to estimate the area under a curve. What is the formula for the area of a trapezoid with bases of length b_1 and b_2 and height h?

(f) Working by hand, estimate the area under $f(x) = x^2$ on $[0, 3]$ using three subintervals and three corresponding trapezoids. What is the error in this approximation? How does it compare to the errors you calculated in (d)?

Activity 5.6.2. In this activity, we explore the relationships among the errors generated by left, right, midpoint, and trapezoid approximations to the definite integral $\int_1^2 \frac{1}{x^2}\,dx$.

(a) Use the First FTC to evaluate $\int_1^2 \frac{1}{x^2}\,dx$ exactly.

(b) Use appropriate computing technology to compute the following approximations for $\int_1^2 \frac{1}{x^2}\,dx$: T_4, M_4, T_8, and M_8.

(c) Let the *error* that results from an approximation be the approximation's value minus the exact value of the definite integral. For instance, if we let $E_{T,4}$ represent the error that results from using the trapezoid rule with 4 subintervals to estimate the integral, we have

$$E_{T,4} = T_4 - \int_1^2 \frac{1}{x^2}\,dx.$$

Similarly, we compute the error of the midpoint rule approximation with 8 subintervals by the formula

$$E_{M,8} = M_8 - \int_1^2 \frac{1}{x^2}\,dx.$$

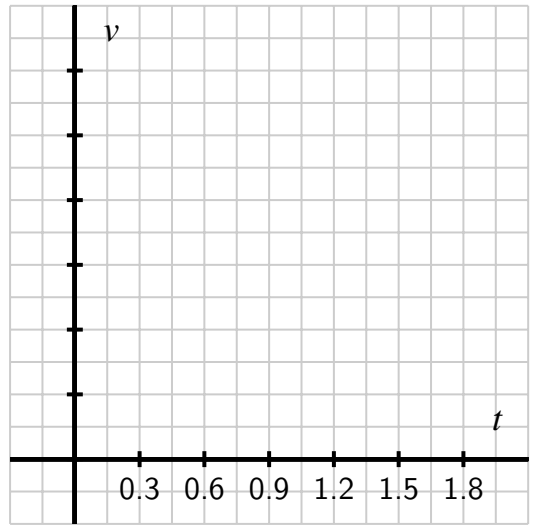
Based on your work in (a) and (b) above, compute $E_{T,4}$, $E_{T,8}$, $E_{M,4}$, $E_{M,8}$.

(d) Which rule consistently over-estimates the exact value of the definite integral? Which rule consistently under-estimates the definite integral?

(e) What behavior(s) of the function $f(x) = \frac{1}{x^2}$ lead to your observations in (d)?

Activity 5.6.3. A car traveling along a straight road is braking and its velocity is measured at several different points in time, as given in the following table. Assume that v is continuous, always decreasing, and always concave down, as is suggested by the data.

seconds, t	Velocity in ft/sec, $v(t)$
0	100
0.3	99
0.6	96
0.9	90
1.2	80
1.5	50
1.8	0



(a) Plot the given data on the set of axes provided in the figure, with time on the horizontal axis and the velocity on the vertical axis.

(b) What definite integral will give you the exact distance the car traveled on $[0, 1.8]$?

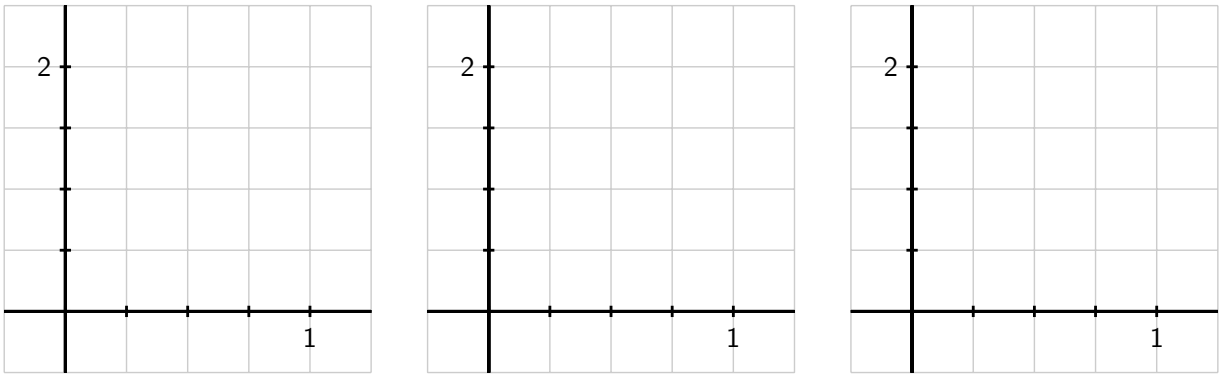
(c) Estimate the total distance traveled on $[0, 1.8]$ by computing L_3, R_3, and T_3. Which of these underestimates the true distance traveled?

(d) Estimate the total distance traveled on $[0, 1.8]$ by computing M_3. Is this an over- or under-estimate? Why?

(e) Using your results from (c) and (d), improve your estimate further by using Simpson's Rule.

(f) What is your best estimate of the average velocity of the car on $[0, 1.8]$? Why? What are the units on this quantity?

Activity 5.6.4. Consider the functions $f(x) = 2 - x^2$, $g(x) = 2 - x^3$, and $h(x) = 2 - x^4$, all on the interval $[0, 1]$. For each of the questions that require a numerical answer in what follows, write your answer exactly in fraction form.



(a) On the three sets of axes in the provided figure, sketch a graph of each function on the interval $[0, 1]$, and compute L_1 and R_1 for each. What do you observe?

(b) Compute M_1 for each function to approximate $\int_0^1 f(x)\,dx$, $\int_0^1 g(x)\,dx$, and $\int_0^1 h(x)\,dx$, respectively.

(c) Compute T_1 for each of the three functions, and hence compute S_2 for each of the three functions.

(d) Evaluate each of the integrals $\int_0^1 f(x)\,dx$, $\int_0^1 g(x)\,dx$, and $\int_0^1 h(x)\,dx$ exactly using the First FTC.

(e) For each of the three functions f, g, and h, compare the results of L_1, R_1, M_1, T_1, and S_2 to the true value of the corresponding definite integral. What patterns do you observe?

CHAPTER 6

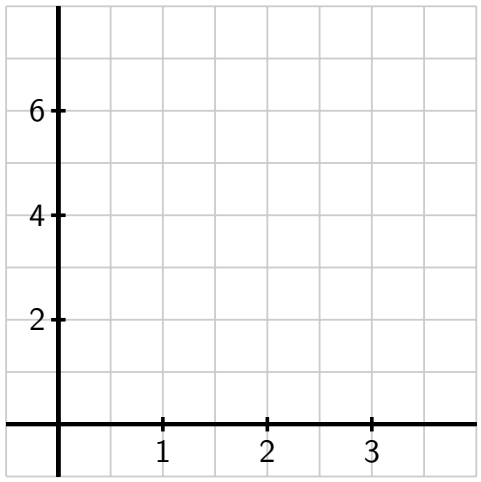
Using Definite Integrals

6.1 Using definite integrals to find area and length

Preview Activity 6.1.1. Consider the functions given by $f(x) = 5 - (x-1)^2$ and $g(x) = 4 - x$.

(a) Use algebra to find the points where the graphs of f and g intersect.

(b) Sketch an accurate graph of f and g on the axes provided, labeling the curves by name and the intersection points with ordered pairs.



CC BY SA

(c) Find and evaluate exactly an integral expression that represents the area between $y = f(x)$ and the x-axis on the interval between the intersection points of f and g.

(d) Find and evaluate exactly an integral expression that represents the area between $y = g(x)$ and the x-axis on the interval between the intersection points of f and g.

(e) What is the exact area between f and g between their intersection points? Why?

Activity 6.1.2. In each of the following situations, our goal is to determine the area of the region described. For each region, (i) determine the intersection points of the curves, (ii) sketch the region whose area is being found, (iii) draw and label a representative slice, and (iv) state the area of the representative slice. Then, state a definite integral whose value is the exact area of the region, and evaluate the integral to find the numeric value of the region's area.

(a) The finite region bounded by $y = \sqrt{x}$ and $y = \frac{1}{4}x$.

(b) The finite region bounded by $y = 12 - 2x^2$ and $y = x^2 - 8$.

(c) The area bounded by the y-axis, $f(x) = \cos(x)$, and $g(x) = \sin(x)$, where we consider the region formed by the first positive value of x for which f and g intersect.

(d) The finite regions between the curves $y = x^3 - x$ and $y = x^2$.

Activity 6.1.3. In each of the following situations, our goal is to determine the area of the region described. For each region, (i) determine the intersection points of the curves, (ii) sketch the region whose area is being found, (iii) draw and label a representative slice, and (iv) state the area of the representative slice. Then, state a definite integral whose value is the exact area of the region, and evaluate the integral to find the numeric value of the region's area. *Note well:* At the step where you draw a representative slice, you need to make a choice about whether to slice vertically or horizontally.

(a) The finite region bounded by $x = y^2$ and $x = 6 - 2y^2$.

(b) The finite region bounded by $x = 1 - y^2$ and $x = 2 - 2y^2$.

(c) The area bounded by the x-axis, $y = x^2$, and $y = 2 - x$.

(d) The finite regions between the curves $x = y^2 - 2y$ and $y = x$.

Activity 6.1.4. Each of the following questions involves the arc length along a curve.

(a) Use the definition and appropriate computational technology to determine the arc length along $y = x^2$ from $x = -1$ to $x = 1$.

(b) Find the arc length of $y = \sqrt{4 - x^2}$ on the interval $-2 \leq x \leq 2$. Find this value in two different ways: (a) by using a definite integral, and (b) by using a familiar property of the curve.

(c) Determine the arc length of $y = xe^{3x}$ on the interval $[0, 1]$.

(d) Will the integrals that arise calculating arc length typically be ones that we can evaluate exactly using the First FTC, or ones that we need to approximate? Why?

(e) A moving particle is traveling along the curve given by $y = f(x) = 0.1x^2 + 1$, and does so at a constant rate of 7 cm/sec, where both x and y are measured in cm (that is, the curve $y = f(x)$ is the path along which the object actually travels; the curve is not a "position function"). Find the position of the particle when $t = 4$ sec, assuming that when $t = 0$, the particle's location is $(0, f(0))$.

6.2 Using definite integrals to find volume

Preview Activity 6.2.1. Consider a circular cone of radius 3 and height 5, which we view horizontally as pictured in Figure 6.2.1. Our goal in this activity is to use a definite integral to determine the volume of the cone.

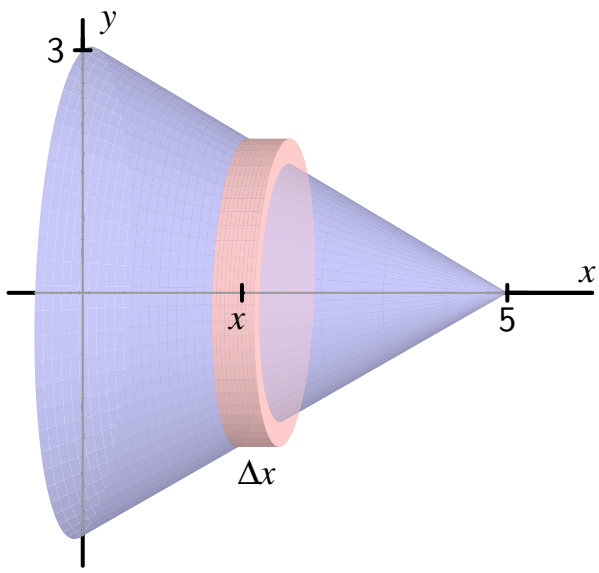


Figure 6.2.1 The circular cone described in Preview Activity 6.2.1

(a) Find a formula for the linear function $y = f(x)$ that is pictured in Figure 6.2.1.

(b) For the representative slice of thickness Δx that is located horizontally at a location x (somewhere between $x = 0$ and $x = 5$), what is the radius of the representative slice? Note that the radius depends on the value of x.

(c) What is the volume of the representative slice you found in (b)?

(d) What definite integral will sum the volumes of the thin slices across the full horizontal span of the cone? What is the exact value of this definite integral?

(e) Compare the result of your work in (d) to the volume of the cone that comes from using the formula $V_{\text{cone}} = \frac{1}{3}\pi r^2 h$.

Activity 6.2.2. In each of the following questions, draw a careful, labeled sketch of the region described, as well as the resulting solid that results from revolving the region about the stated axis. In addition, draw a representative slice and state the volume of that slice, along with a definite integral whose value is the volume of the entire solid. It is not necessary to evaluate the integrals you find.

(a) The region S bounded by the x-axis, the curve $y = \sqrt{x}$, and the line $x = 4$; revolve S about the x-axis.

(b) The region S bounded by the y-axis, the curve $y = \sqrt{x}$, and the line $y = 2$; revolve S about the x-axis.

(c) The finite region S bounded by the curves $y = \sqrt{x}$ and $y = x^3$; revolve S about the x-axis.

(d) The finite region S bounded by the curves $y = 2x^2 + 1$ and $y = x^2 + 4$; revolve S about the x-axis.

(e) The region S bounded by the y-axis, the curve $y = \sqrt{x}$, and the line $y = 2$; revolve S about the y-axis. How is this problem different from the one posed in part (b)?

Activity 6.2.3. In each of the following questions, draw a careful, labeled sketch of the region described, as well as the resulting solid that results from revolving the region about the stated axis. In addition, draw a representative slice and state the volume of that slice, along with a definite integral whose value is the volume of the entire solid. It is not necessary to evaluate the integrals you find.

(a) The region S bounded by the y-axis, the curve $y = \sqrt{x}$, and the line $y = 2$; revolve S about the y-axis.

(b) The region S bounded by the x-axis, the curve $y = \sqrt{x}$, and the line $x = 4$; revolve S about the y-axis.

(c) The finite region S in the first quadrant bounded by the curves $y = 2x$ and $y = x^3$; revolve S about the x-axis.

(d) The finite region S in the first quadrant bounded by the curves $y = 2x$ and $y = x^3$; revolve S about the y-axis.

(e) The finite region S bounded by the curves $x = (y-1)^2$ and $y = x - 1$; revolve S about the y-axis

Activity 6.2.4. In each of the following questions, draw a careful, labeled sketch of the region described, as well as the resulting solid that results from revolving the region about the stated axis. In addition, draw a representative slice and state the volume of that slice, along with a definite integral whose value is the volume of the entire solid. It is not necessary to evaluate the integrals you find. For each prompt, use the finite region S in the first quadrant bounded by the curves $y = 2x$ and $y = x^3$.

(a) Revolve S about the line $y = -2$.

(b) Revolve S about the line $y = 4$.

(c) Revolve S about the line $x = -1$.

(d) Revolve S about the line $x = 5$.

6.3 Density, mass, and center of mass

Preview Activity 6.3.1. In each of the following scenarios, we consider the distribution of a quantity along an axis.

(a) Suppose that the function $c(x) = 200+100e^{-0.1x}$ models the density of traffic on a straight road, measured in cars per mile, where x is number of miles east of a major interchange, and consider the definite integral $\int_0^2 (200 + 100e^{-0.1x})\, dx$.

i. What are the units on the product $c(x) \cdot \Delta x$?

ii. What are the units on the definite integral and its Riemann sum approximation given by

$$\int_0^2 c(x)\, dx \approx \sum_{i=1}^{n} c(x_i)\Delta x?$$

iii. Evaluate the definite integral $\int_0^2 c(x)\, dx = \int_0^2 \left(200 + 100e^{-0.1x}\right)\, dx$ and write one sentence to explain the meaning of the value you find.

(b) On a 6 foot long shelf filled with books, the function B models the distribution of the weight of the books, in pounds per inch, where x is the number of inches from the left end of the bookshelf. Let $B(x)$ be given by the rule $B(x) = 0.5 + \frac{1}{(x+1)^2}$.

i. What are the units on the product $B(x) \cdot \Delta x$?

ii. What are the units on the definite integral and its Riemann sum approximation given by

$$\int_{12}^{36} B(x)\, dx \approx \sum_{i=1}^{n} B(x_i)\Delta x?$$

iii. Evaluate the definite integral $\int_0^{72} B(x)\, dx = \int_0^{72} \left(0.5 + \frac{1}{(x+1)^2}\right)\, dx$ and write one sentence to explain the meaning of the value you find.

Activity 6.3.2. Consider the following situations in which mass is distributed in a non-constant manner.

(a) Suppose that a thin rod with constant cross-sectional area of 1 cm^2 has its mass distributed according to the density function $\rho(x) = 2e^{-0.2x}$, where x is the distance in cm from the left end of the rod, and the units on $\rho(x)$ are g/cm. If the rod is 10 cm long, determine the exact mass of the rod.

(b) Consider the cone that has a base of radius 4 m and a height of 5 m. Picture the cone lying horizontally with the center of its base at the origin and think of the cone as a solid of revolution.

i. Write and evaluate a definite integral whose value is the volume of the cone.

ii. Next, suppose that the cone has uniform density of 800 kg/m^3. What is the mass of the solid cone?

iii. Now suppose that the cone's density is not uniform, but rather that the cone is most dense at its base. In particular, assume that the density of the cone is uniform across cross sections parallel to its base, but that in each such cross section that is a distance x units from the origin, the density of the cross section is given by the function $\rho(x) = 400 + \frac{200}{1+x^2}$, measured in kg/m^3. Determine and evaluate a definite integral whose value is the mass of this cone of non-uniform density. Do so by first thinking about the mass of a given slice of the cone x units away from the base; remember that in such a slice, the density will be *essentially constant*.

(c) Let a thin rod of constant cross-sectional area 1 cm^2 and length 12 cm have its mass be distributed according to the density function $\rho(x) = \frac{1}{25}(x - 15)^2$, measured in g/cm. Find the exact location z at which to cut the bar so that the two pieces will each have identical mass.

Activity 6.3.3. For quantities of equal weight, such as two children on a teeter-totter, the balancing point is found by taking the average of their locations. When the weights of the quantities differ, we use a weighted average of their respective locations to find the balancing point.

(a) Suppose that a shelf is 6 feet long, with its left end situated at $x = 0$. If one book of weight 1 lb is placed at $x_1 = 0$, and another book of weight 1 lb is placed at $x_2 = 6$, what is the location of $\overline{x}$, the point at which the shelf would (theoretically) balance on a fulcrum?

(b) Now, say that we place four books on the shelf, each weighing 1 lb: at $x_1 = 0$, at $x_2 = 2$, at $x_3 = 4$, and at $x_4 = 6$. Find $\overline{x}$, the balancing point of the shelf.

(c) How does $\overline{x}$ change if we change the location of the third book? Say the locations of the 1-lb books are $x_1 = 0$, $x_2 = 2$, $x_3 = 3$, and $x_4 = 6$.

(d) Next, suppose that we place four books on the shelf, but of varying weights: at $x_1 = 0$ a 2-lb book, at $x_2 = 2$ a 3-lb book, at $x_3 = 4$ a 1-lb book, and at $x_4 = 6$ a 1-lb book. Use a weighted average of the locations to find $\overline{x}$, the balancing point of the shelf. How does the balancing point in this scenario compare to that found in (b)?

(e) What happens if we change the location of one of the books? Say that we keep everything the same in (d), except that $x_3 = 5$. How does $\overline{x}$ change?

(f) What happens if we change the weight of one of the books? Say that we keep everything the same in (d), except that the book at $x_3 = 4$ now weighs 2 lbs. How does $\overline{x}$ change?

(g) Experiment with a couple of different scenarios of your choosing where you move one of the books to the left, or you decrease the weight of one of the books.

(h) Write a couple of sentences to explain how adjusting the location of one of the books or the weight of one of the books affects the location of the balancing point of the shelf. Think carefully here about how your changes should be considered relative to the location of the balancing point $\overline{x}$ of the current scenario.

Activity 6.3.4. Consider a thin bar of length 20 cm whose density is distributed according to the function $\rho(x) = 4 + 0.1x$, where $x = 0$ represents the left end of the bar. Assume that ρ is measured in g/cm and x is measured in cm.

(a) Find the total mass, M, of the bar.

(b) Without doing any calculations, do you expect the center of mass of the bar to be equal to 10, less than 10, or greater than 10? Why?

(c) Compute $\overline{x}$, the exact center of mass of the bar.

(d) What is the average density of the bar?

(e) Now consider a different density function, given by $p(x) = 4e^{0.020732x}$, also for a bar of length 20 cm whose left end is at $x = 0$. Plot both $\rho(x)$ and $p(x)$ on the same axes. Without doing any calculations, which bar do you expect to have the greater center of mass? Why?

(f) Compute the exact center of mass of the bar described in (e) whose density function is $p(x) = 4e^{0.020732x}$. Check the result against the prediction you made in (e).

6.4 Physics applications: work, force, and pressure

Preview Activity 6.4.1. A bucket is being lifted from the bottom of a 50-foot deep well; its weight (including the water), B, in pounds at a height h feet above the water is given by the function $B(h)$. When the bucket leaves the water, the bucket and water together weigh $B(0) = 20$ pounds, and when the bucket reaches the top of the well, $B(50) = 12$ pounds. Assume that the bucket loses water at a constant rate (as a function of height, h) throughout its journey from the bottom to the top of the well.

(a) Find a formula for $B(h)$.

(b) Compute the value of the product $B(5)\Delta h$, where $\Delta h = 2$ feet. Include units on your answer. Explain why this product represents the approximate work it took to move the bucket of water from $h = 5$ to $h = 7$.

(c) Is the value in (b) an over- or under-estimate of the actual amount of work it took to move the bucket from $h = 5$ to $h = 7$? Why?

(d) Compute the value of the product $B(22)\Delta h$, where $\Delta h = 0.25$ feet. Include units on your answer. What is the meaning of the value you found?

(e) More generally, what does the quantity $W_{\text{slice}} = B(h)\Delta h$ measure for a given value of h and a small positive value of Δh?

(f) Evaluate the definite integral $\int_0^{50} B(h)\,dh$. What is the meaning of the value you find? Why?

Activity 6.4.2. Consider the following situations in which a varying force accomplishes work.

(a) Suppose that a heavy rope hangs over the side of a cliff. The rope is 200 feet long and weighs 0.3 pounds per foot; initially the rope is fully extended. How much work is required to haul in the entire length of the rope? (Hint: set up a function $F(h)$ whose value is the weight of the rope remaining over the cliff after h feet have been hauled in.)

(b) A leaky bucket is being hauled up from a 100 foot deep well. When lifted from the water, the bucket and water together weigh 40 pounds. As the bucket is being hauled upward at a constant rate, the bucket leaks water at a constant rate so that it is losing weight at a rate of 0.1 pounds per foot. What function $B(h)$ tells the weight of the bucket after the bucket has been lifted h feet? What is the total amount of work accomplished in lifting the bucket to the top of the well?

(c) Now suppose that the bucket in (b) does not leak at a constant rate, but rather that its weight at a height h feet above the water is given by $B(h) = 25 + 15e^{-0.05h}$. What is the total work required to lift the bucket 100 feet? What is the average force exerted on the bucket on the interval $h = 0$ to $h = 100$?

(d) From physics, *Hooke's Law* for springs states that the amount of force required to hold a spring that is compressed (or extended) to a particular length is proportionate to the distance the spring is compressed (or extended) from its natural length. That is, the force to compress (or extend) a spring x units from its natural length is $F(x) = kx$ for some constant k (which is called the *spring constant*.) For springs, we choose to measure the force in pounds and the distance the spring is compressed in feet. Suppose that a force of 5 pounds extends a particular spring 4 inches (1/3 foot) beyond its natural length.

i. Use the given fact that $F(1/3) = 5$ to find the spring constant k.

ii. Find the work done to extend the spring from its natural length to 1 foot beyond its natural length.

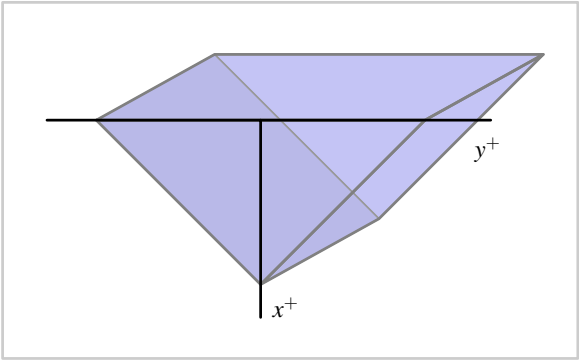
iii. Find the work required to extend the spring from 1 foot beyond its natural length to 1.5 feet beyond its natural length.

Activity 6.4.3. In each of the following situations, determine the total work required to accomplish the described task. In parts (b) and (c), a key step is to find a formula for a function that describes the curve that forms the side boundary of the tank.

(a) Consider a vertical cylindrical tank of radius 2 meters and depth 6 meters. Suppose the tank is filled with 4 meters of water of mass density 1000 kg/m^3, and the top 1 meter of water is pumped over the top of the tank.

(b) Consider a hemispherical tank with a radius of 10 feet. Suppose that the tank is full to a depth of 7 feet with water of weight density 62.4 $pounds/ft^3$, and the top 5 feet of water are pumped out of the tank to a tanker truck whose height is 5 feet above the top of the tank.

(c) Consider a trough with triangular ends, as pictured in the following figure where the tank is 10 feet long, the top is 5 feet wide, and the tank is 4 feet deep. Say that the trough is full to within 1 foot of the top with water of weight density 62.4 pounds/ft^3, and a pump is used to empty the tank until the water remaining in the tank is 1 foot deep.

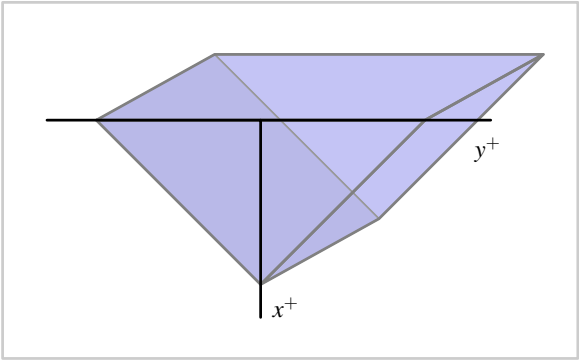


Activity 6.4.4. In each of the following situations, determine the total force exerted by water against the surface that is described.

(a) Consider a rectangular dam that is 100 feet wide and 50 feet tall, and suppose that water presses against the dam all the way to the top.

(b) Consider a semicircular dam with a radius of 30 feet. Suppose that the water rises to within 10 feet of the top of the dam.

(c) Consider a trough with triangular ends, as pictured in the following figure where the tank is 10 feet long, the top is 5 feet wide, and the tank is 4 feet deep. Say that the trough is full to within 1 foot of the top with water of weight density 62.4 pounds/ft^3. How much force does the water exert against one of the triangular ends?



6.5 Improper integrals

Preview Activity 6.5.1. A company with a large customer base has a call center that receives thousands of calls a day. After studying the data that represents how long callers wait for assistance, they find that the function $p(t) = 0.25e^{-0.25t}$ models the time customers wait in the following way: the fraction of customers who wait between $t = a$ and $t = b$ minutes is given by

$$\int_a^b p(t)\,dt.$$

Use this information to answer the following questions.

(a) Determine the fraction of callers who wait between 5 and 10 minutes.

(b) Determine the fraction of callers who wait between 10 and 20 minutes.

(c) Next, let's study the fraction who wait up to a certain number of minutes:

i. What is the fraction of callers who wait between 0 and 5 minutes?

ii. What is the fraction of callers who wait between 0 and 10 minutes?

iii. Between 0 and 15 minutes? Between 0 and 20?

(d) Let $F(b)$ represent the fraction of callers who wait between 0 and b minutes. Find a formula for $F(b)$ that involves a definite integral, and then use the First FTC to find a formula for $F(b)$ that does not involve a definite integral.

(e) What is the value of $\lim_{b\to\infty} F(b)$? What is the meaning of this limit in the context of the problem?

Activity 6.5.2. In this activity we explore the improper integrals $\int_1^\infty \frac{1}{x}\,dx$ and $\int_1^\infty \frac{1}{x^{3/2}}\,dx$.

(a) First we investigate $\int_1^\infty \frac{1}{x}\,dx$.

i. Use the First FTC to determine the exact values of $\int_1^{10} \frac{1}{x}\,dx$, $\int_1^{1000} \frac{1}{x}\,dx$, and $\int_1^{100000} \frac{1}{x}\,dx$. Then, use your computational device to compute a decimal approximation of each result.

ii. Use the First FTC to evaluate the definite integral $\int_1^b \frac{1}{x}\,dx$ (which results in an expression that depends on b).

iii. Now, use your work from (ii.) to evaluate the limit given by

$$\lim_{b\to\infty} \int_1^b \frac{1}{x}\,dx.$$

(b) Next, we investigate $\int_1^\infty \frac{1}{x^{3/2}}\,dx$.

i. Use the First FTC to determine the exact values of $\int_1^{10} \frac{1}{x^{3/2}}\,dx$, $\int_1^{1000} \frac{1}{x^{3/2}}\,dx$, and $\int_1^{100000} \frac{1}{x^{3/2}}\,dx$. Then, use your calculator to compute a decimal approximation of each result.

ii. Use the First FTC to evaluate the definite integral $\int_1^b \frac{1}{x^{3/2}}\,dx$ (which results in an expression that depends on b).

iii. Now, use your work from (ii.) to evaluate the limit given by

$$\lim_{b\to\infty} \int_1^b \frac{1}{x^{3/2}}\,dx.$$

(c) Plot the functions $y = \frac{1}{x}$ and $y = \frac{1}{x^{3/2}}$ on the same coordinate axes for the values $x = 0 \ldots 10$. How would you compare their behavior as x increases without bound? What is similar? What is different?

(d) How would you characterize the value of $\int_1^\infty \frac{1}{x}\,dx$? of $\int_1^\infty \frac{1}{x^{3/2}}\,dx$? What does this tell us about the respective areas bounded by these two curves for $x \geq 1$?

Activity 6.5.3. Determine whether each of the following improper integrals converges or diverges. For each integral that converges, find its exact value.

(a) $\int_1^\infty \frac{1}{x^2}\,dx$

(b) $\int_0^\infty e^{-x/4}\,dx$

(c) $\int_2^\infty \frac{9}{(x+5)^{2/3}}\,dx$

(d) $\int_4^\infty \frac{3}{(x+2)^{5/4}}\,dx$

(e) $\int_0^\infty xe^{-x/4}\,dx$

(f) $\int_1^\infty \frac{1}{x^p}\,dx$, where p is a positive real number

Activity 6.5.4. For each of the following definite integrals, decide whether the integral is improper or not. If the integral is proper, evaluate it using the First FTC. If the integral is improper, determine whether or not the integral converges or diverges; if the integral converges, find its exact value.

(a) $\int_0^1 \frac{1}{x^{1/3}}\, dx$

(b) $\int_0^2 e^{-x}\, dx$

(c) $\int_1^4 \frac{1}{\sqrt{4-x}}\, dx$

(d) $\int_{-2}^{2} \frac{1}{x^2}\, dx$

(e) $\int_{0}^{\pi/2} \tan(x)\, dx$

(f) $\int_{0}^{1} \frac{1}{\sqrt{1-x^2}}\, dx$

CHAPTER 7

Differential Equations

7.1 An introduction to differential equations

Preview Activity 7.1.1. The position of a moving object is given by the function $s(t)$, where s is measured in feet and t in seconds. We determine that the object's velocity is $v(t) = 4t + 1$ feet per second.

(a) How much does the position change over the time interval $[0, 4]$?

(b) Does this give you enough information to determine $s(4)$, the position at time $t = 4$? If so, what is $s(4)$? If not, what additional information would you need to know to determine $s(4)$?

(c) Suppose you are told that the object's initial position $s(0) = 7$. Determine $s(2)$, the object's position 2 seconds later.

(d) If you are told instead that the object's initial position is $s(0) = 3$, what is $s(2)$?

(e) If we only know the velocity $v(t) = 4t + 1$, is it possible that the object's position at all times is $s(t) = 2t^2 + t - 4$? Explain how you know.

(f) Are there other possibilities for $s(t)$? If so, what are they?

(g) If, in addition to knowing the velocity function is $v(t) = 4t + 1$, we know the initial position $s(0)$, how many possibilities are there for $s(t)$?

Activity 7.1.2. Express the following statements as differential equations. In each case, you will need to introduce notation to describe the important quantities in the statement, so be sure to clearly state what your notation means.

(a) The population of a town grows continuously at an annual rate of 1.25%.

(b) A radioactive sample loses mass at a rate of 5.6% of its mass every day.

(c) You have a bank account that continuously earns 4% interest every year. At the same time, you withdraw money continually from the account at the rate of $1000 per year.

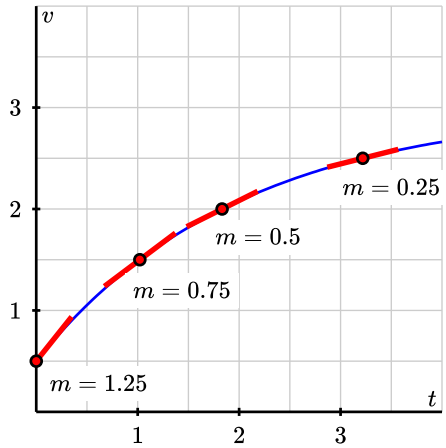
(d) A cup of hot chocolate is sitting in a 70° room. The temperature of the hot chocolate cools continuously by 10% of the difference between the hot chocolate's temperature and the room temperature every minute.

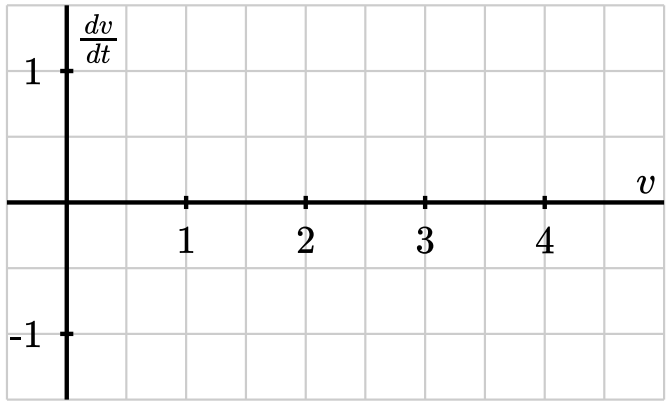
(e) A can of cold soda is sitting in a 70° room. The temperature of the soda warms continuously at the rate of 10% of the difference between the soda's temperature and the room's temperature every minute.

Activity 7.1.3. In this activity, we consider information about the velocity of objects falling through the atmosphere: a skydiver in part (a), followed by a meteorite in part (b).

(a) Begin with the skydiver's velocity v given by the graph at left below, which shows how v changes as t changes. Use this graph to find the rate of change dv/dt at the points where the velocity is $v = 0.5, 1.5, 2.0,$ and 2.5. Note particularly that several key slopes are provided for you on the given graph.

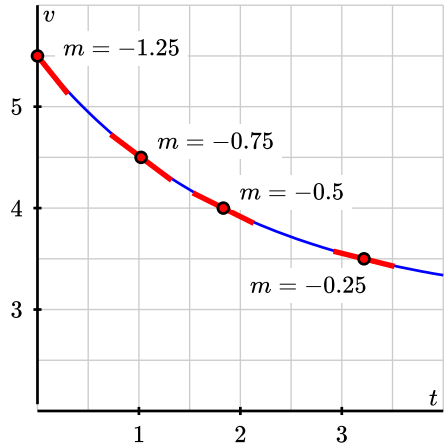
Then, on the axes provided, plot the values of the derivative dv/dt as a function of *velocity*. Think carefully about this: we are focusing on how we can connect the velocity v to the resulting value of dv/dt. For example, in the velocity graph at left, we observe from one of the points that when $v = 2$, $dv/dt = 0.5$.





(b) Next we're going to consider a falling meteorite's velocity.

While the skydiver fell faster and faster, the meteorite falls slower and slower through the atmosphere, as shown in the graph at right. Note the scale on the vertical axis: since the velocity is always greater than $v = 3$, we only show v values from 3 to 6. Reasoning similarly to part (a), use the given graph of the meteorite's velocity to find the rate of change dv/dt at the points where the velocity is $v = 3.5, 4.0, 4.5,$ and 5.0. Plot the appropriate resulting points on the same axes used in part (a).



(c) You should find that all of the points you plotted on the axes in (a) lie on a line. Remember that these points show how dv/dt depends on v. Write the equation of this line, being careful to use proper notation for the quantities on the horizontal and vertical axes.

(d) The relationship you just found in (c) is a differential equation. Write a complete sentence that explains its meaning.

(e) Use the differential equation you found in (c) to determine the values of the velocity for which the velocity increases. Write a sentence to explain your thinking.

(f) Similarly, use the differential equation to determine the values of the velocity for which the velocity decreases. Again, explain your thinking.

(g) Finally, determine the value(s) of the velocity for which the velocity remains constant. What do these values mean in the overall context of this activity?

Activity 7.1.4. Consider the differential equation

$$\frac{dv}{dt} = 1.5 - 0.5v.$$

Determine, with justification, whether or not each of the following functions is a solution to the differential equation.

(a) $v(t) = 1.5t - 0.25t^2$.

(b) $v(t) = 3 + 2e^{-0.5t}$.

(c) $v(t) = 3$.

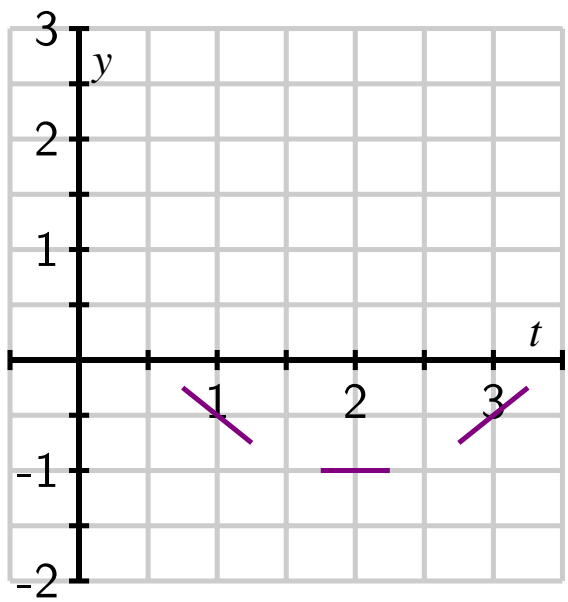
(d) $v(t) = 3 + Ce^{-0.5t}$ where C is any constant.

7.2 Qualitative behavior of solutions to differential equations

Preview Activity 7.2.1. Let's consider the initial value problem

$$\frac{dy}{dt} = t - 2, \quad y(0) = 1.$$

(a) Use the differential equation to find the slope of the tangent line to the solution $y(t)$ at $t = 0$. Then use the initial value to find the equation of the tangent line at $t = 0$. Sketch this tangent line over the interval $-0.25 \leq t \leq 0.25$ on the axes provided below.



(b) Also shown in the figure in (a) are the tangent lines to the solution $y(t)$ at the points $t = 1, 2$, and 3 (we will see how to find these later). Use the graph to measure the slope of each tangent line and verify that each agrees with the value specified by the differential equation.

(c) Using the tangent lines from (a) and (b) as a guide, sketch a graph of the solution $y(t)$ over the interval $0 \leq t \leq 3$ so that the lines are tangent to the graph of $y(t)$.

(d) Use the Fundamental Theorem of Calculus to find $y(t)$, the solution to original initial value problem,

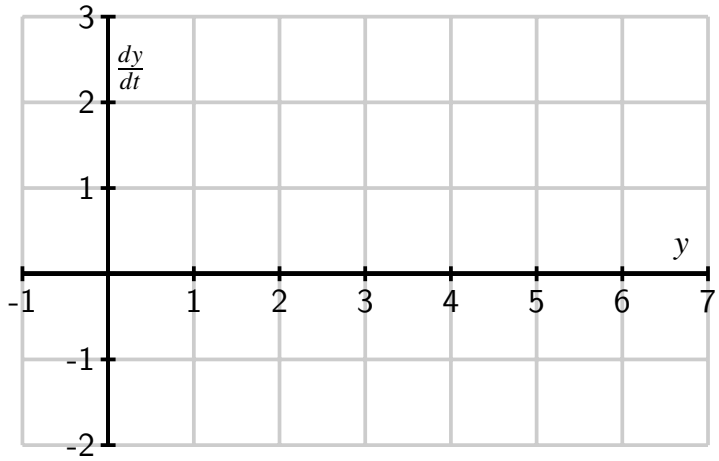
$$\frac{dy}{dt} = t - 2, \quad y(0) = 1.$$

(e) Graph the solution you found in (d) on the axes provided in (a), and compare it to the sketch you made using the tangent lines.

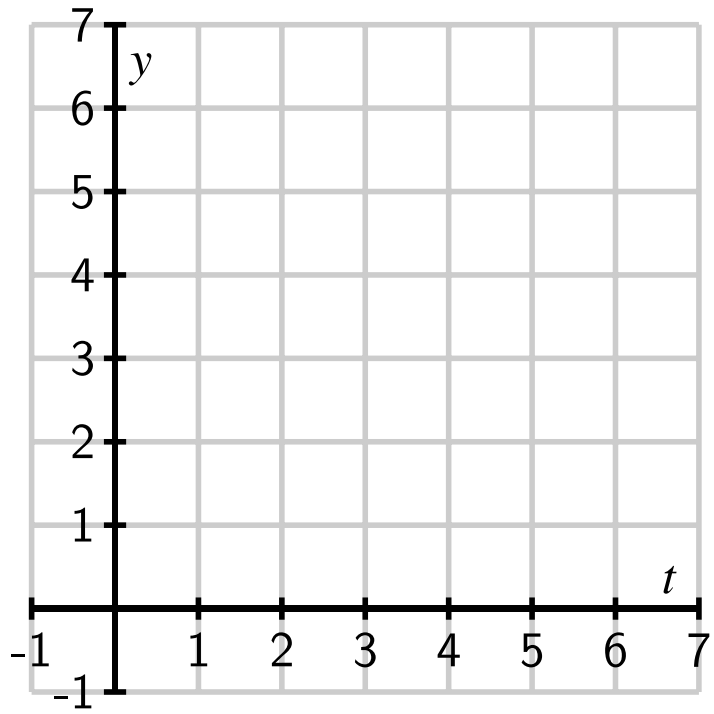
Activity 7.2.2. Consider the autonomous differential equation

$$\frac{dy}{dt} = -\frac{1}{2}(y-4).$$

(a) Make a plot of $\frac{dy}{dt}$ versus y on the axes provided below. Looking at the graph, for what values of y does y increase and for what values of y does y decrease?



(b) Next, sketch the slope field for this differential equation on the axes provided below.



(c) Use your work in (b) to sketch on the same axes solutions that satisfy $y(0) = 0$, $y(0) = 2$, $y(0) = 4$ and $y(0) = 6$.

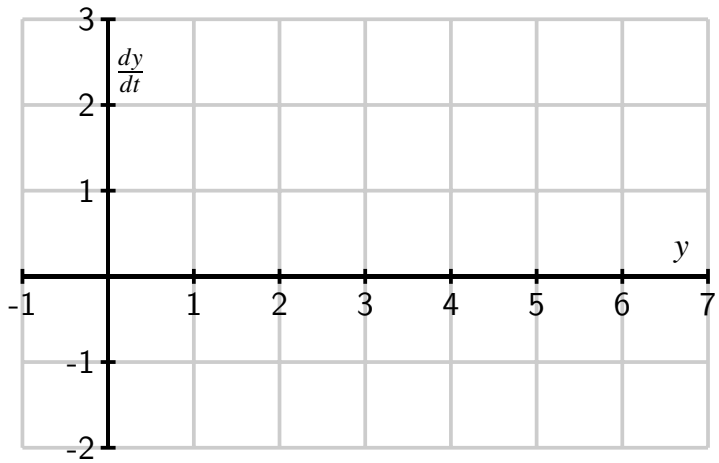
(d) Verify that $y(t) = 4 + 2e^{-t/2}$ is a solution to the given differential equation with the initial value $y(0) = 6$. Compare its graph to the one you sketched in (c).

(e) What is special about the solution where $y(0) = 4$?

Activity 7.2.3. Consider the autonomous differential equation

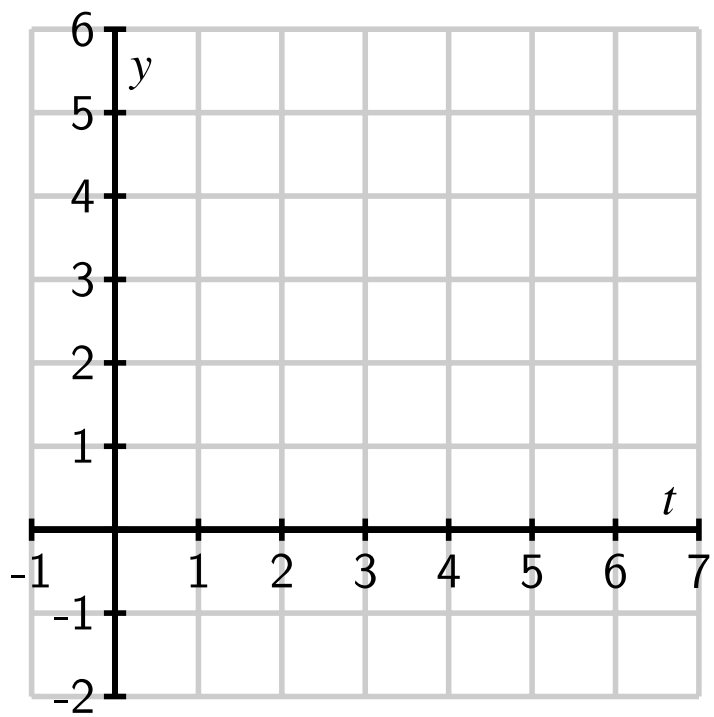
$$\frac{dy}{dt} = -\frac{1}{2}y(y-4).$$

(a) Make a plot of $\frac{dy}{dt}$ versus y on the axes provided. Looking at the graph, for what values of y does y increase and for what values of y does y decrease?



(b) Identify any equilibrium solutions of the given differential equation.

(c) Now sketch the slope field for the given differential equation on the axes provided below.

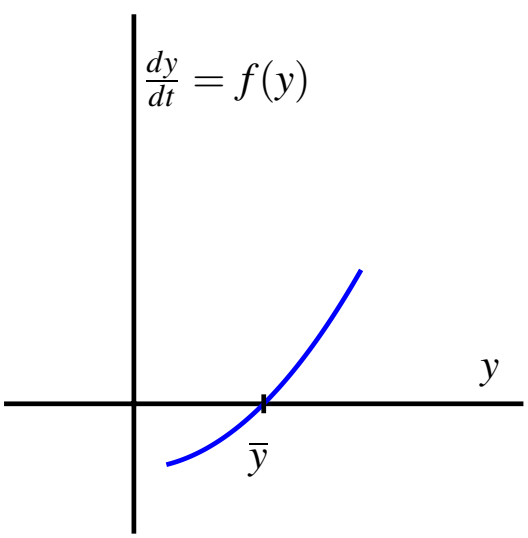


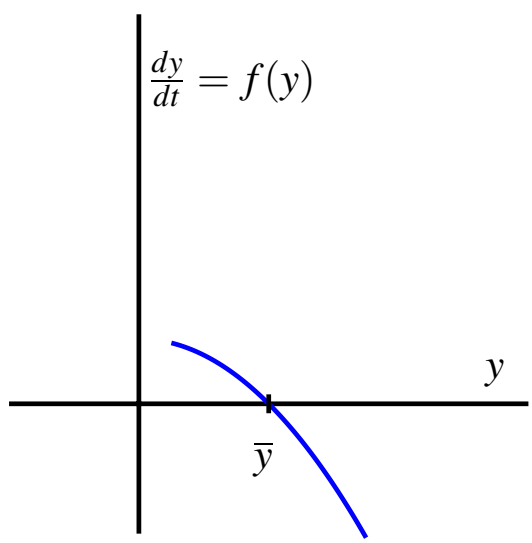
(d) On the slope field in (c), sketch the solutions to the given differential equation that correspond to initial values $y(0) = -1, 0, 1, \ldots, 5$.

(e) An equilibrium solution $\overline{y}$ is called *stable* if nearby solutions converge to $\overline{y}$. This means that if the initial condition varies slightly from $\overline{y}$, then $\lim_{t\to\infty} y(t) = \overline{y}$. Conversely, an equilibrium solution $\overline{y}$ is called *unstable* if nearby solutions are pushed away from $\overline{y}$. Using your work above, classify the equilibrium solutions you found in (b) as either stable or unstable.

(f) Suppose that $y(t)$ describes the population of a species of living organisms and that the initial value $y(0)$ is positive. What can you say about the eventual fate of this population?

(g) Now consider a general autonomous differential equation of the form $dy/dt = f(y)$. Remember that an equilibrium solution $\overline{y}$ satisfies $f(\overline{y}) = 0$. If we graph $dy/dt = f(y)$ as a function of y, for which of the differential equations represented below is $\overline{y}$ a stable equilibrium and for which is $\overline{y}$ unstable? Why?





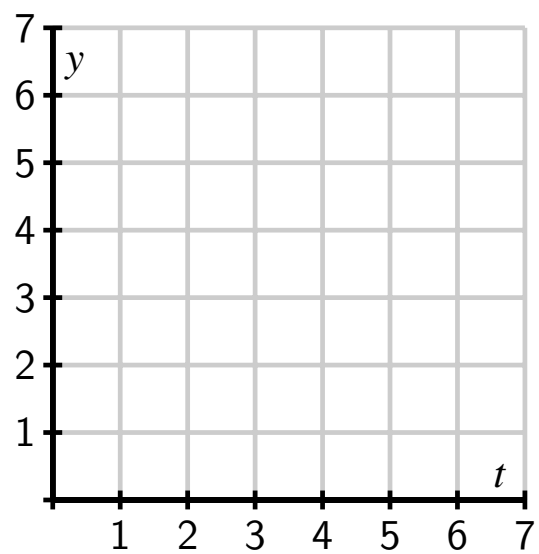
7.3 Euler's method

Preview Activity 7.3.1. Consider the initial value problem

$$\frac{dy}{dt} = \frac{1}{2}(y+1), \; y(0) = 0.$$

(a) Use the differential equation to find the slope of the tangent line to the solution $y(t)$ at $t = 0$. Then use the given initial value to find the equation of the tangent line at $t = 0$.

(b) Sketch the tangent line you found in (a) on the axes provided on the interval $0 \leq t \leq 2$ and use the tangent line to approximate $y(2)$, the value of the solution at $t = 2$.



(c) Assuming that your approximation for $y(2)$ from (b) is the actual value of $y(2)$, use the differential equation to find the slope of the tangent line to $y(t)$ at $t = 2$ (i.e., at the point $(2, y(2))$). Then, write the equation of the tangent line at $t = 2$.

(d) Add a sketch of this tangent line from (c) on the interval $2 \leq t \leq 4$ to your plot in (b); use this new tangent line to approximate $y(4)$, the value of the solution at $t = 4$.

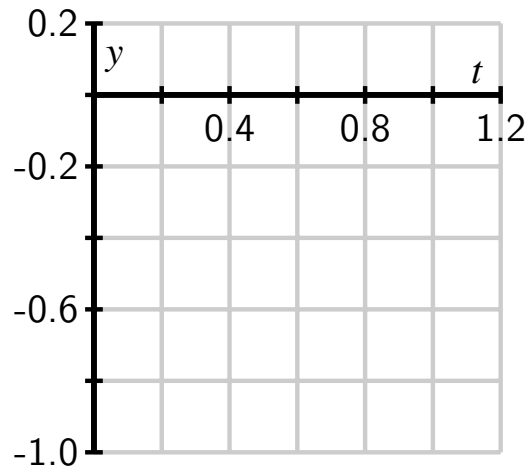
(e) Repeat the same step to find an approximation for $y(6)$.

Activity 7.3.2. Consider the initial value problem

$$\frac{dy}{dt} = 2t - 1, \; y(0) = 0$$

(a) Use Euler's method with $\Delta t = 0.2$ to approximate the solution at $t_i = 0.2, 0.4, 0.6, 0.8$, and 1.0. Record your work in the following table, and sketch the points (t_i, y_i) on the axes provided.

t_i	y_i	dy/dt	Δy
0.0000	0.0000		
0.2000			
0.4000			
0.6000			
0.8000			
1.0000			



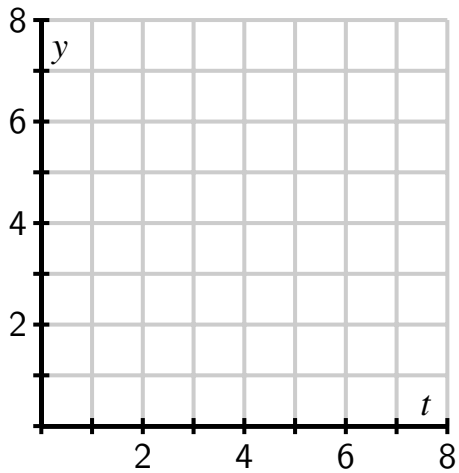
(b) Find the exact solution to the original initial value problem and use this function to find the error in your approximation at each one of the points t_i.

(c) How would your computations differ if the initial value was $y(0) = 1$? What does this mean about different solutions to this differential equation?

(d) Explain why the value y_5 generated by Euler's method for this initial value problem produces the same value as a left Riemann sum for the definite integral $\int_0^1 (2t - 1)\, dt$.

Activity 7.3.3. Consider the differential equation $\frac{dy}{dt} = 6y - y^2$.

(a) Sketch the slope field for this differential equation on the axes provided.

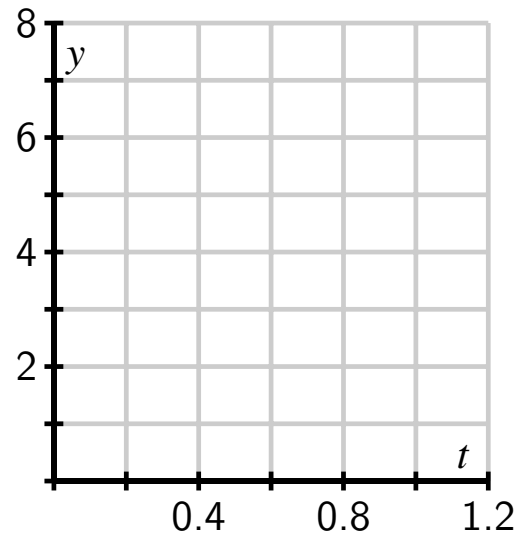


(b) Identify any equilibrium solutions and determine whether they are stable or unstable.

(c) What is the long-term behavior of the solution that satisfies the initial value $y(0) = 1$?

(d) Using the initial value $y(0) = 1$, use Euler's method with $\Delta t = 0.2$ to approximate the solution at $t_i = 0.2, 0.4, 0.6, 0.8$, and 1.0. Record your results in the table and sketch the corresponding points (t_i, y_i) on the axes provided. Note the different horizontal scale on the axes here compared to the axes above.

t_i	y_i	dy/dt	Δy
0.0	1.0000		
0.2			
0.4			
0.6			
0.8			
1.0			



(e) What happens if we apply Euler's method to approximate the solution with $y(0) = 6$?

7.4 Separable differential equations

Preview Activity 7.4.1. In this preview activity, we explore whether certain differential equations are separable or not, and then revisit some key ideas from earlier work in integral calculus.

(a) Which of the following differential equations are separable? If the equation is separable, write the equation in the revised form $g(y)\frac{dy}{dt} = h(t)$.

i. $\dfrac{dy}{dt} = -3y$.

ii. $\dfrac{dy}{dt} = ty - y$.

iii. $\dfrac{dy}{dt} = t + 1$.

iv. $\dfrac{dy}{dt} = t^2 - y^2$.

(b) Explain why any autonomous differential equation is guaranteed to be separable.

(c) Why do we include the term "$+C$" in the expression

$$\int x\,dx = \frac{x^2}{2} + C?$$

(d) Suppose we know that a certain function f satisfies the equation

$$\int f'(x)\,dx = \int x\,dx.$$

What can you conclude about f?

Activity 7.4.2. Suppose that the population of a town is growing continuously at an annual rate of 3% per year.

(a) Let $P(t)$ be the population of the town in year t. Write a differential equation that describes the annual growth rate.

(b) Find the solutions of this differential equation.

(c) If you know that the town's population in year 0 is 10,000, find the population $P(t)$.

(d) How long does it take for the population to double? This time is called the *doubling time*.

(e) Working more generally, find the doubling time if the annual growth rate is k times the population.

Activity 7.4.3. Suppose that a cup of coffee is initially at a temperature of 105° F and is placed in a 75° F room. If T is the temperature of the coffee in degrees Fahrenheit at time t in minutes, Newton's law of cooling says that

$$\frac{dT}{dt} = -k(T - 75),$$

where k is a constant of proportionality.

(a) Suppose you measure that the coffee is cooling at one degree per minute at the time the coffee is brought into the room. Use the differential equation to determine the value of the constant k.

(b) Find all the solutions of this differential equation.

(c) What happens to all the solutions as $t \to \infty$? Explain how this agrees with your intuition.

(d) What is the temperature of the cup of coffee after 20 minutes?

(e) How long does it take for the coffee to cool to 80°?

Activity 7.4.4. Solve each of the following differential equations or initial value problems.

(a) $\frac{dy}{dt} - (2-t)y = 2 - t$

(b) $\frac{1}{t}\frac{dy}{dt} = e^{t^2-2y}$

(c) $y' = 2y + 2,\ y(0) = 2$

(d) $y' = 2y^2,\ y(-1) = 2$

(e) $\frac{dy}{dt} = \frac{-2ty}{t^2+1},\ y(0) = 4$

7.5 Modeling with differential equations

Preview Activity 7.5.1. Any time that the rate of change of a quantity is related to the amount of a quantity, a differential equation naturally arises. In the following prompts, we encouner two such scenarios; for each, we want to develop a differential equation whose solution is the quantity of interest.

(a) Suppose you have a bank account in which money grows at an annual rate of 3%.

If you have \$10,000 in the account, at what rate is your money growing?

(b) Suppose that you are withdrawing money from the account at \$1,000 per year. What is the rate of change in the amount of money in the account that these withdrawals induce? What are the units on this rate of change?

(c) For a bank account with \$$P$ that earns 3% annual interest and from which you withdraw \$1,000 per year, what differential equation does P satisfy?

(d) Now consider a different scenario. Suppose that a water tank holds 100 gallons and that a salty solution, which contains 20 grams of salt in every gallon, enters the tank at 2 gallons per minute.

How many grams of salt enter the tank each minute?

(e) Suppose that initially there are 300 grams of salt in the tank. How many grams of salt are in each gallon at this point in time?

(f) Finally, let $A(t)$ represent the amount of salt in the tank at time t. Suppose that evenly mixed solution is pumped out of the tank at the rate of 2 gallons per minute. How much salt leaves the tank each minute?

(g) What is the total rate of change in the amount of salt in the tank? Use your observations in the preceding questions (that involve the rates at which salt enters and leaves the tank) to write a differential equation satisfied by $A(t)$.

Activity 7.5.2. Suppose you have a bank account that grows by 5% every year. Let $A(t)$ be the amount of money in the account in year t.

(a) What is the rate of change of A with respect to t?

(b) Suppose that you are also withdrawing \$10,000 per year. Write a differential equation that expresses the total rate of change of A.

(c) Sketch a slope field for this differential equation, find any equilibrium solutions, and identify them as either stable or unstable. Write a sentence or two that describes the significance of the stability of the equilibrium solution.

(d) Suppose that you initially deposit \$100,000 into the account. How long does it take for you to deplete the account?

(e) What is the smallest amount of money you would need to have in the account to guarantee that you never deplete the money in the account?

(f) If your initial deposit is $300,000, how much could you withdraw every year without depleting the account?

Activity 7.5.3. A dose of morphine is absorbed from the bloodstream of a patient at a rate proportional to the amount in the bloodstream.

(a) Write a differential equation for $M(t)$, the amount of morphine in the patient's bloodstream, using k as the constant proportionality.

(b) Assuming that the initial dose of morphine is M_0, solve the initial value problem to find $M(t)$. Use the fact that the half-life for the absorption of morphine is two hours to find the constant k.

(c) Suppose that a patient is given morphine intravenously at the rate of 3 milligrams per hour. Write a differential equation that combines the intravenous administration of morphine with the body's natural absorption.

(d) Find any equilibrium solutions and determine their stability.

(e) Assuming that there is initially no morphine in the patient's bloodstream, solve the initial value problem to determine $M(t)$. What happens to $M(t)$ after a very long time?

(f) To what rate should a doctor reduce the intravenous rate so that there is eventually 7 milligrams of morphine in the patient's bloodstream?

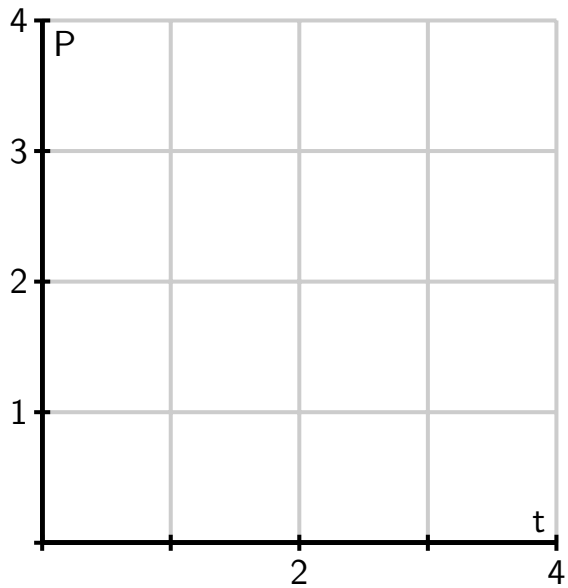
7.6 Population growth and the logistic equation

Preview Activity 7.6.1. Recall that one model for population growth states that a population grows at a rate proportional to its size. In symbols: $\frac{dP}{dt} = kP$.

(a) We begin with the differential equation

$$\frac{dP}{dt} = \frac{1}{2}P.$$

Sketch a slope field as well as a few typical solutions on the axes provided.



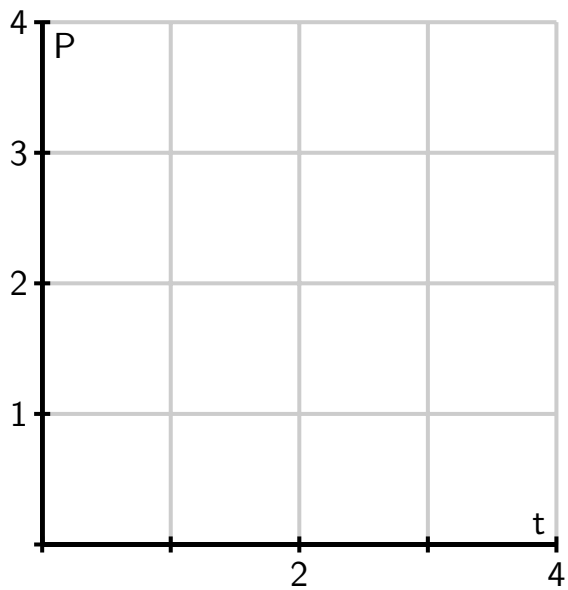
(b) Find all equilibrium solutions of the equation $\frac{dP}{dt} = \frac{1}{2}P$ and classify each as stable or unstable.

(c) If $P(0)$ is positive, describe the long-term behavior of the solution to $\frac{dP}{dt} = \frac{1}{2}P$.

(d) Let's now consider a modified differential equation given by

$$\frac{dP}{dt} = \frac{1}{2}P(3 - P).$$

As before, sketch a slope field as well as a few typical solutions on the following axes provided.



(e) Find any equilibrium solutions and classify them as stable or unstable.

(f) If $P(0)$ is positive, describe the long-term behavior of the solution.

Activity 7.6.2. Using some recent population data for planet Earth, we want to model the growth of the population over time.

Table 7.6.1 Some recent population data (in billions) for planet Earth.

Year	1998	1999	2000	2001	2002	2005	2006	2007	2008	2009	2010
Pop	5.932	6.008	6.084	6.159	6.234	6.456	6.531	6.606	6.681	6.756	6.831

Our first model will be based on the following assumption:

The rate of change of the population is proportional to the population.

On the face of it, this seems pretty reasonable. When there is a relatively small number of people, there will be fewer births and deaths so the rate of change will be small. When there is a larger number of people, there will be more births and deaths so we expect a larger rate of change. If $P(t)$ is the population t years after the year 2000, we may express this assumption as

$$\frac{dP}{dt} = kP$$

where k is a constant of proportionality.

(a) Use the data in the table to estimate the derivative $P'(0)$ using a central difference. Assume that $t = 0$ corresponds to the year 2000.

(b) What is the population $P(0)$?

(c) Use your results from (a) and (b) to estimate the constant of proportionality k in the differential equation.

(d) Now that we know the value of k, we have the initial value problem

$$\frac{dP}{dt} = kP, \ P(0) = 6.084.$$

Find the solution to this initial value problem.

(e) What does your solution predict for the population in the year 2010? Is this close to the actual population given in the table?

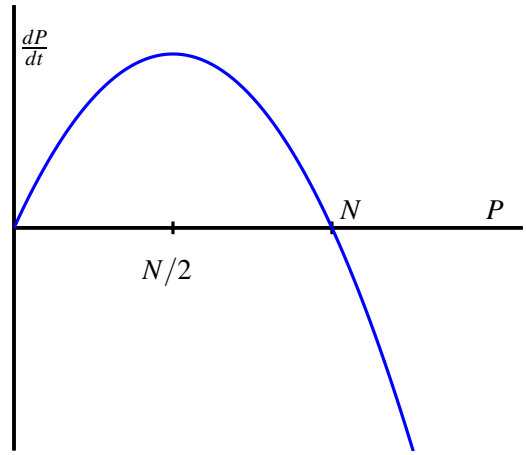
(f) When does your solution predict that the population will reach 12 billion?

(g) What does your solution predict for the population in the year 2500?

Activity 7.6.3. Consider the logistic equation

$$\frac{dP}{dt} = kP(N - P)$$

with the graph of $\frac{dP}{dt}$ vs. P shown below.



(a) At what value of P is the rate of change greatest?

(b) Consider the model for the earth's population that we recently created: $\frac{dP}{dt} = P(0.025 - 0.002P)$. At what value of P is the rate of change greatest? How does that compare to the population in recent years?

(c) According to the model we developed (recall we found that $P = \frac{12.5}{1.0546e^{-0.025t}+1}$), what will the population be in the year 2100?

(d) According to the model we developed, when will the population reach 9 billion?

(e) Now consider the general solution to the general logistic initial value problem that we found, given by

$$P(t) = \frac{N}{\left(\frac{N-P_0}{P_0}\right) e^{-kNt} + 1}.$$

Verify algebraically that $P(0) = P_0$ and that $\lim_{t\to\infty} P(t) = N$.

CHAPTER 8

Taylor Polynomials and Taylor Series

8.1 Extending local linearization

Preview Activity 8.1.1. Consider the function $f(x) = e^x$ near $a = 0$. We know that $f'(x) = e^x$, so $f'(0) = 1$; along with the fact that $f(0) = 1$, it follows that the tangent line approximation is

$$L(x) = f(0) + f'(0)(x - 0) = 1 + 1(x - 0) = 1 + x.$$

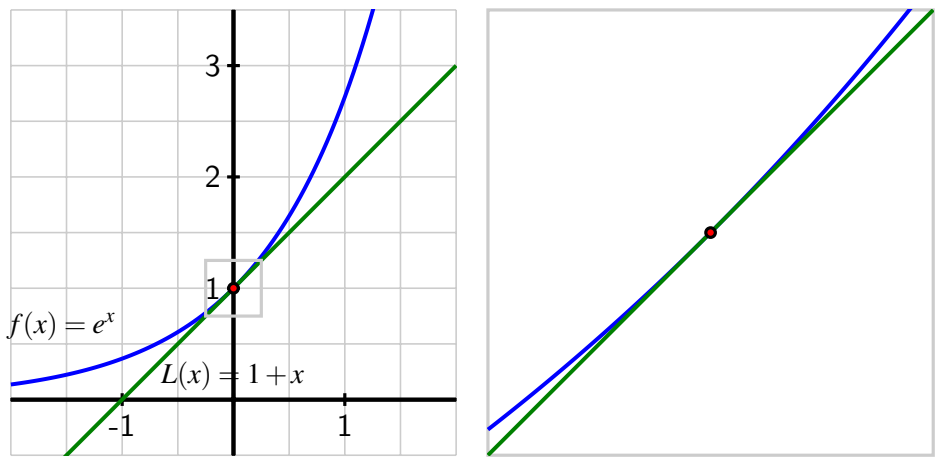


Figure 8.1.1 The function $f(x) = e^x$ and its tangent line $L(x) = 1 + x$ near the point $(0, f(0))$.

(a) Build a spreadsheet that computes the difference between $f(x)$ and $L(x)$ for x-values between -1 and 1, spaced 0.1 units apart. ***Note:*** we will revisit this spreadsheet for additional computations in Activity 8.1.4, so it would be ideal if you save your work for later reference.

Your spreadsheet should start like the one shown in Table 8.1.2.

Table 8.1.2 Comparing $f(x) = e^x$ and $L(x) = 1 + x$ near $a = 0$.

Δx	x	$f(x) = e^x$	$L(x) = 1 + x$
0.1	-1.0	0.36788	0.0
0.1	-0.9	0.40657	0.1
0.1	-0.8	0.44933	0.2

(b) Next, add a column to your spreadsheet that computes $|f(x) - L(x)|$.

How do the differences between $f(x)$ and $L(x)$ change as you move across the interval from $x = -1$ to $x = 1$?

(c) For approximately what values of x is it true that $|f(x) - L(x)| < 0.1$?

(d) Notice that the curvature in $f(x) = e^x$ is what makes the linear approximation $L(x)$ lose accuracy. What kind of simple (non-exponential) function might do a better job approximating e^x than a linear one?

Activity 8.1.2. Let $f(x) = e^x$ and $T_2(x) = b_0 + b_1x + b_2x^2$. We seek numerical values for the constants b_0, b_1, and b_2 so that $f(0) = T_2(0)$, $f'(0) = T_2'(0)$, and $f''(0) = T_2''(0)$.

(a) Note that since b_0, b_1, and b_2 are constants, if we take the derivative of the quadratic function T_2 using the sum and constant multiple rules, it follows that $T_2'(x) = b_1 + 2b_2x$.

What is $T_2''(x)$?

(b) Recall that $f(x) = e^x$. Determine $f'(x)$ and $f''(x)$.

(c) Enter the formulas you've determined for $f'(x)$, $f''(x)$, and $T_2''(x)$ to fill in the blanks below.

$f(x) = e^x$	$T_2(x) = b_0 + b_1x + b_2x^2$
$f'(x) =$	$T_2'(x) = b_1 + 2b_2x$
$f''(x) =$	$T_2''(x) =$

(d) Next, observe that since $T_2(x) = b_0 + b_1x + b_2x^2$, it follows that $T_2(0) = b_0$. Reason similarly to determine the values of $T_2'(0)$ and $T_2''(0)$, as well as those of $f(0)$, $f'(0)$, and $f''(0)$ and enter these values appropriately in the blanks below.

$f(0) =$	$T_2(0) = b_0$
$f'(0) =$	$T_2'(0) =$
$f''(0) =$	$T_2''(0) =$

(e) Now, recall that we want the function values, first derivative values, and second derivative values of f and T_2 to match at $a = 0$. What does $T_2(0) = f(0)$ tell us about the value of b_0, and what is its value? What does $T_2'(0) = f'(0)$ imply the value of b_1 is? How can we reason similarly to find b_2?

(f) Having now determined the numerical values of b_0, b_1, and b_2, use appropriate computing technology to plot the function $T_2(x) = b_0 + b_1 x + b_2 x^2$ along with $f(x) = e^x$ and $T_1(x) = 1 + x$ in the same window as that shown in Figure 8.1.3.

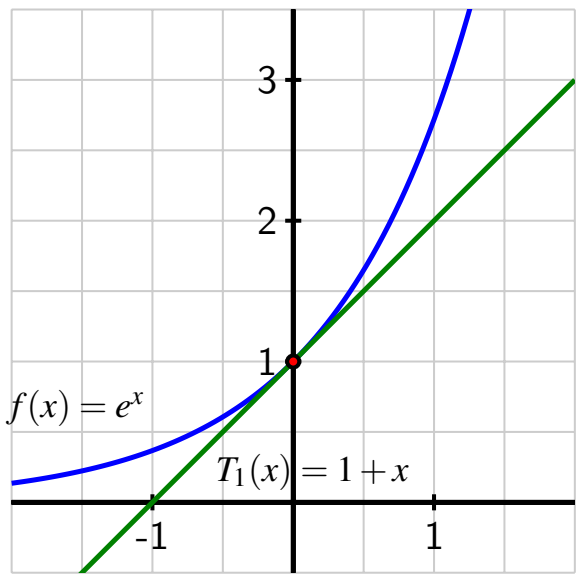


Figure 8.1.3 The function $f(x) = e^x$ and its tangent line $T_1(x) = 1 + x$ near the point $(0, f(0))$.

What do you notice? For about which values of x is $|f(x) - T_2(x)| < 0.1$?

Activity 8.1.3. Let $f(x) = e^x$ and $T_3(x) = c_0 + c_1x + c_2x^2 + c_3x^3$.

(a) By computing the third derivative of $f(x)$ and the second and third derivatives of $T_3(x)$ and evaluating the relevant functions at $x = 0$, fill in the blanks below.

$f(x) = e^x$	$T_3(x) = c_0 + c_1x + c_2x^2 + c_3x^3$
$f'(x) = e^x$	$T_3'(x) = c_1 + 2c_2x + 3c_3x^2$
$f''(x) = e^x$	$T_3''(x) =$
$f'''(x) =$	$T_3'''(x) =$
$f(0) = 1$	$T_3(0) =$
$f'(0) = 1$	$T_3'(0) =$
$f''(0) = 1$	$T_3''(0) =$
$f'''(0) =$	$T_3'''(0) =$

(b) Next, recall that we want f and T_3 to share the same function and derivative values at $a = 0$ up to and including the third derivative. For instance, one of the four needed equations is $T_3'(0) = f'(0)$. Use the four equations your work in the preceding question to determine the values of c_0, c_1, c_2, and c_3.

(c) Having now determined the numerical values of c_0, c_1, c_2, and c_3, use appropriate computational technology to plot the function $T_3(x) = c_0 + c_1x + c_2x^2 + c_3x^3$ along with $f(x) = e^x$, $T_1(x) = 1 + x$, and $T_2(x) = 1 + x + \frac{1}{2}x^2$ in the same window as shown in Figure 8.1.4.

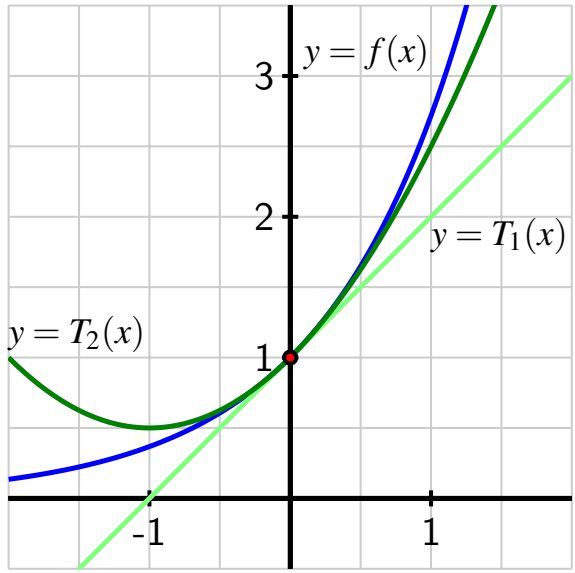


Figure 8.1.4 The function $f(x) = e^x$, its tangent line $T_1(x) = 1 + x$, and the quadratic approximation $T_2(x) = 1 + x + \frac{1}{2}x^2$ near the point $(0, f(0))$.

What do you notice? For approximately which values of x is $|f(x) - T_3(x)| < 0.1$?

(d) What if we wanted a degree-4 polynomial approximation to $f(x) = e^x$ near $a = 0$? Based on the patterns you've observed in T_1, T_2, and T_3, conjecture values for the constants $d_0, \ldots, d_4$ for a function T_4 of the form

$$T_4(x) = d_0 + d_1x + d_2x^2 + d_3x^3 + d_4x^4$$

that satisfies $T_4(0) = f(0)$, $T_4'(0) = f'(0)$, $\ldots$, $T_4^{(4)}(0) = f^{(4)}(0)$. Add this function T_4 to your plot in part (c) that includes $f(x)$ and the lower-degree polynomial approximations. What do you notice?

Activity 8.1.4. We continue to work with $f(x) = e^x$ and the four approximations of degree 1, 2, 3, and 4 given by $T_1(x)$, $T_2(x)$, $T_3(x)$, and $T_4(x)$.

(a) In Preview Activity 8.1.1, we built a spreadsheet that computed the differences between $f(x)$ and $T_1(x)$ for x-values between -1 and 1, spaced 0.1 units apart. Your spreadsheet started like the one shown in the table in Preview Activity 8.1.1.

Next, we build an updated version of this spreadsheet that computes similar differences between f and the three higher degree approximations we have found. In particular, we now want to have columns for Δx, x, $f(x)$, $T_1(x)$, $T_2(x)$, $T_3(x)$, and $T_4(x)$, plus the absolute differences $|f(x) - T_1(x)|$, $|f(x) - T_2(x)|$, $|f(x) - T_3(x)|$, and $|f(x) - T_4(x)|$. ***Hint:*** when building your entries, note that you can think of $T_2(x)$ as $T_2(x) = T_1(x) + \frac{1}{2}x^2$, and similarly view $T_3(x)$ as "$T_2(x)$ plus one more term".

Include at least 5 digits of accuracy beyond the decimal. The first seven columns of your spreadsheet might start like this:

Table 8.1.5 Comparing $f(x) = e^x$ and its degree 1, 2, 3, 4 and approximations near $a = 0$.

Δx	x	$f(x)$	$T_1(x)$	$T_2(x)$	$T_3(x)$	$T_4(x)$
0.1	-1.0	0.36787	0.00000	0.50000	0.33333	0.37500
0.1	-0.9	0.40657	0.10000	0.50500	0.38350	0.41083

The next four columns of your spreadsheet should begin as follows:

Table 8.1.6 The absolute error between $f(x) = e^x$ and its degree 1, 2, 3, and 4 approximations at $x = -1$ and $x = -0.9$.

$\lvert f(x) - T_1(x)\rvert$	$\lvert f(x) - T_2(x)\rvert$	$\lvert f(x) - T_3(x)\rvert$	$\lvert f(x) - T_4(x)\rvert$
0.36787	0.13212	0.03454	0.00712
0.30657	0.09843	0.02307	0.00426

(b) We call the value of $|f(x) - T_2(x)|$ the **absolute error of the quadratic approximation of f at the value x**. What is the absolute error of the quadratic approximation at $x = -1$? at $x = 1$?

(c) What is the absolute error of the cubic (degree 3) approximation, $T_3(x)$, at $x = -1$? at $x = 1$?

(d) What is the absolute error of the quartic (degree 4) approximation, $T_4(x)$, at $x = -1$? at $x = 1$?

(e) Study your spreadsheet for trends that you notice as the value of x changes or the degree n of the approximation changes. What are your observations?

(f) Investigate the errors in the various approximations for a wider interval of x-values. For example, you might consider starting at $x = -2$ with $\Delta x = 0.2$. What do you notice?

8.2 Taylor polynomials

Preview Activity 8.2.1. Let $f(x) = \sin(x)$ and let $T_3(x) = c_0 + c_1x + c_2x^2 + c_3x^3$. Our goal is to find the values of $c_0, \ldots, c_3$ that make the sine function and its derivative values agree with those of the cubic polynomial T_3 at $a = 0$ and to study the resulting degree 3 approximation of the sine function.

(a) As in previous work, the derivatives of $T_3(x)$ and their respective values at $a = 0$ are those shown below. Compute the first three derivatives of $f(x) = \sin(x)$ and evaluate them at $a = 0$ accordingly, recording your results in the blanks provided below.

$f(x) = \sin(x)$	$T_3(x) = c_0 + c_1x + c_2x^2 + c_3x^3$
$f'(x) =$	$T_3'(x) = c_1 + 2c_2x + 3c_3x^2$
$f''(x) =$	$T_3''(x) = 2c_2 + 6c_3x$
$f'''(x) =$	$T_3'''(x) = 6c_3$

$f(0) =$	$T_3(0) = c_0$
$f'(0) =$	$T_3'(0) = c_1$
$f''(0) =$	$T_3''(0) = 2c_2$
$f'''(0) =$	$T_3'''(0) = 6c_3$

(b) Now, set $T_3(0) = f(0)$, $T_3'(0) = f'(0)$, $T_3''(0) = f''(0)$, and $T_3'''(0) = f'''(0)$. What are the resulting values of $c_0, \ldots, c_3$? What is the resulting formula for $T_3(x)$?

(c) Recall that the tangent line approximation T_1 to $f(x) = \sin(x)$ at $a = 0$ is $T_1(x) = x$. Use appropriate computing technology to plot the cubic approximation $T_3(x)$ you found in (b) along with $f(x)$ and $T_1(x)$ in the same window shown in Figure 8.2.1.

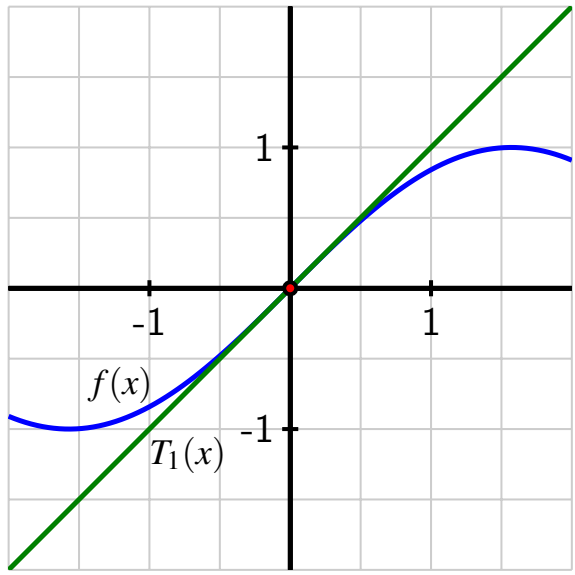


Figure 8.2.1 The function $f(x) = \sin(x)$ and its tangent line approximation $T_1(x) = x$ near the point $(0, f(0))$.

(d) What do you observe about the approximation of $f(x)$ by $T_3(x)$ compared to the approximation of $f(x)$ by $T_1(x)$? For example, how do $f(1) - T_1(1)$ and $f(1) - T_3(1)$ compare?

(e) In our work so far, we've seen that raising the degree of polynomial improves the approximation of the original function $f(x)$. If we wanted to find a degree 9 approximation to $f(x) = \sin(x)$, we'd need to find a polynomial whose value and first 9 derivatives match those of the sine function at $a = 0$.

Fill in the blanks below to determine the next six derivatives of $f(x) = \sin(x)$, up to $f^{(9)}(x)$.

$f(x) = \sin(x)$	$f^{(5)}(x) =$
$f'(x) = \cos(x)$	$f^{(6)}(x) =$
$f''(x) = -\sin(x)$	$f^{(7)}(x) =$
$f'''(x) = -\cos(x)$	$f^{(8)}(x) =$
$f^{(4)}(x) =$	$f^{(9)}(x) =$

What pattern do you notice in the derivatives of $f(x) = \sin(x)$?

Activity 8.2.2. Let $f(x) = \cos(x)$. Through the questions that follow, we seek to find the degree n Taylor polynomial for $f(x)$ centered at $a = 0$.

(a) Determine the first 8 derivatives of $f(x) = \cos(x)$ and evaluate each at $a = 0$. Summarize your work by filling in all the blanks below.

$k = 0$	$f(x) = \cos(x)$	$f(0) = \cos(0) = 1$
$k =$	$f'(x) =$	$f'(0) =$
$k =$	$f''(x) =$	$f''(0) =$
$k =$	$f'''(x) =$	$f'''(0) =$
$k =$	$f^{(4)}(x) =$	$f^{(4)}(0) =$
$k =$	$f^{(5)}(x) =$	$f^{(5)}(0) =$
$k =$	$f^{(6)}(x) =$	$f^{(6)}(0) =$
$k =$	$f^{(7)}(x) =$	$f^{(7)}(0) =$
$k =$	$f^{(8)}(x) =$	$f^{(8)}(0) =$

(b) Use your work in (a) along with the fact that the coefficients of the Taylor polynomial are determined by $c_k = \frac{f^{(k)}(0)}{k!}$ to find formulas for $T_2(x)$, $T_4(x)$, $T_6(x)$, and $T_8(x)$.

(c) Based on the patterns you observe in your prior work, what do you expect to be the formula for $T_{10}(x)$?

(d) Use appropriate computing technology to plot $T_4(x)$ and $T_6(x)$ along with $f(x) = \cos(x)$ and $T_2(x)$ in the same window as shown in Figure 8.2.2.

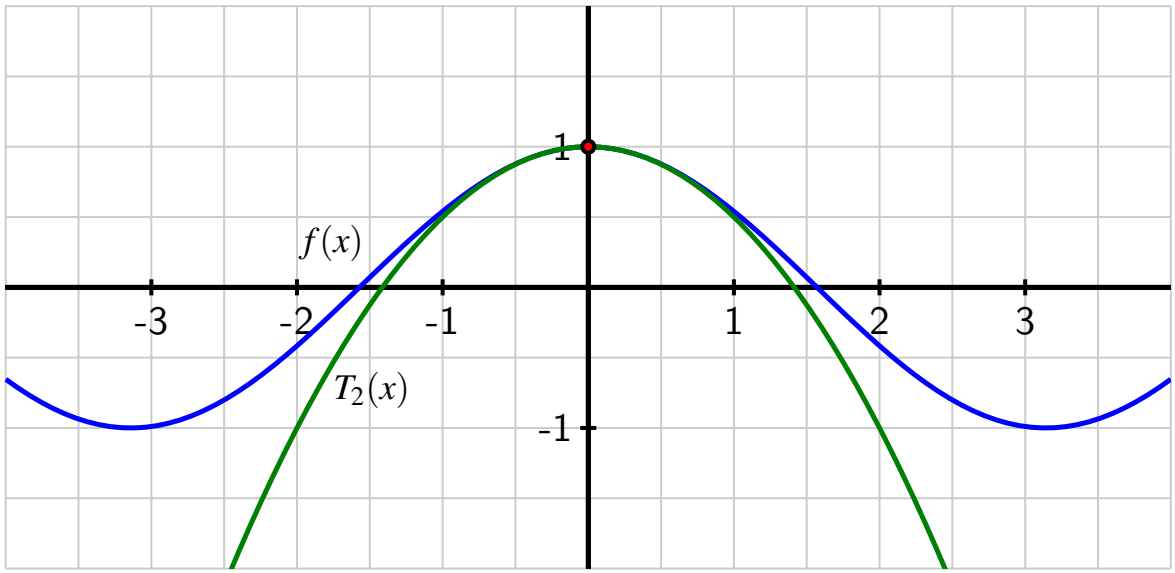


Figure 8.2.2 The function $f(x) = \cos(x)$ and its degree 2 Taylor approximation $T_2(x) = 1 - \frac{1}{2}x^2$ near the point $(0, f(0))$.

What do you notice?

(e) Build a spreadsheet similar to the one in Table 8.1.5 and Table 8.1.6 from Activity 8.1.4, but do so using $\Delta x = 0.2$, a start value of $x = -2$, and the functions $f(x) = \cos(x)$, $T_2(x)$, $T_4(x)$, and $T_6(x)$. The first six columns of your spreadsheet should begin as shown in Table 8.2.3.

Table 8.2.3 Comparing $f(x) = \cos(x)$ and its degree 2, 4, and 6 approximations near $a = 0$.

Δx	x	$f(x)$	$T_2(x)$	$T_4(x)$	$T_6(x)$
0.2	−2.0	−0.41615	−1.00000	−0.33333	−0.42222
0.2	−1.8	−0.22720	−0.62000	−0.18260	−0.22984

and the last three columns of your spreadsheet should begin as follows:

Table 8.2.4 The absolute error between $f(x) = \cos(x)$ and its degree 2, 4, and 6 approximations.

$\lvert f(x) - T_2(x)\rvert$	$\lvert f(x) - T_4(x)\rvert$	$\lvert f(x) - T_6(x)\rvert$
0.58385	0.08281	0.00608
0.39280	0.04460	0.00263

For about what interval of x-values is it true that $|f(x) - T_2(x)| < 0.1$? How does the interval of x-values change if we instead consider where $|f(x) - T_4(x)| < 0.1$? $|f(x) - T_6(x)| < 0.1$?

Activity 8.2.3. Let $f(x) = \ln(x)$, and recall that f is only defined for $x > 0$. As such, we can't consider the tangent line (or any other) approximation at $a = 0$. Instead, we choose to work with an approximation to $f(x) = \ln(x)$ centered at $a = 1$ and will find the degree 4 Taylor polynomial approximation

$$T_4(x) = c_0 + c_1(x-1) + c_2(x-1)^2 + c_3(x-1)^3 + c_4(x-1)^4.$$

(a) Determine $f'(x)$, $f''(x)$, $f'''(x)$, and $f^{(4)}(x)$, and then compute $f'(1)$, $f''(1)$, $f'''(1)$, and $f^{(4)}(1)$. Enter your results in the provided blanks below.

$k = 0$	$f(x) = \ln(x)$	$f(1) = \ln(1) = 0$
$k =$	$f'(x) =$	$f'(1) =$
$k =$	$f''(x) =$	$f''(1) =$
$k =$	$f'''(x) =$	$f'''(1) =$
$k =$	$f^{(4)}(x) =$	$f^{(4)}(1) =$

(b) Use your work in (a) along with the fact that the coefficients of the Taylor polynomial are determined by $c_k = \frac{f^{(k)}(a)}{k!}$ to determine

$$T_4(x) = c_0 + c_1(x-1) + c_2(x-1)^2 + c_3(x-1)^3 + c_4(x-1)^4.$$

(c) Use appropriate technology to plot $f(x) = \ln(x)$, its tangent line, $T_1(x) = x - 1$, and $T_4(x)$ in the same window shown in Figure 8.2.5.

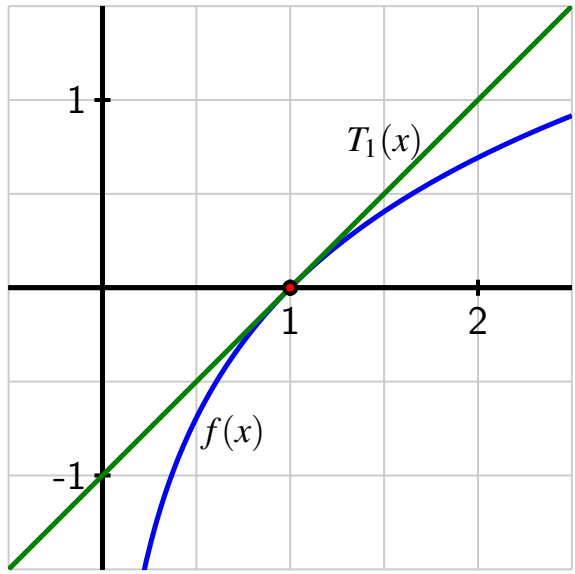


Figure 8.2.5 The function $f(x) = \ln(x)$ and its degree 1 Taylor approximation $T_1(x) = x - 1$ near the point $(1, f(1))$.

What do you notice?

(d) Compute $|f(x) - T_4(x)|$ for several different x values (you might find it helpful to use a slider in *Desmos* in the variable b to experiment with $|f(b) - T_4(b)|$); for approximately what values of x is it true that $|f(x) - T_4(x)| < 0.1$?

(e) Use the patterns you observe in your work in parts (a) and (b) to conjecture formulas for $T_5(x)$ and $T_6(x)$.

For approximately what interval of x-values is it true that $|f(x) - T_5(x)| < 0.1$? What about $|f(x) - T_6(x)| < 0.1$? How is this situation different from what we observed with $f(x) = \cos(x)$ in Activity 8.2.2?

Activity 8.2.4. This activity builds on Activity 8.2.3, and only changes one key thing: the location where the approximation is centered. Again, we let $f(x) = \ln(x)$, and recall that f is only defined for $x > 0$. Here, we choose to work with an approximation centered at $a = 2$, and find the degree 4 Taylor polynomial approximation

$$T_4(x) = c_0 + c_1(x-2) + c_2(x-2)^2 + c_3(x-2)^3 + c_4(x-2)^4.$$

(a) We recall $f'(x)$, $f''(x)$, $f'''(x)$, and $f^{(4)}(x)$ from our work in Activity 8.2.3, and then compute $f'(2)$, $f''(2)$, $f'''(2)$, and $f^{(4)}(2)$. Enter the updated results in the blanks below.

$k = 0$	$f(x) = \ln(x)$	$f(2) =$
$k = 1$	$f'(x) = x^{-1}$	$f'(2) =$
$k = 2$	$f''(x) = (-1)x^{-2}$	$f''(2) =$
$k = 3$	$f'''(x) = (-2)(-1)x^{-3}$	$f'''(2) =$
$k = 4$	$f^{(4)}(x) = (-3)(-2)(-1)x^{-4}$	$f^{(4)}(2) =$

(b) Use your work in (a) along with the fact that the coefficients of the Taylor polynomial are determined by $c_k = \frac{f^{(k)}(a)}{k!}$ to determine $T_4(x) = c_0 + c_1(x-2) + c_2(x-2)^2 + c_3(x-2)^3 + c_4(x-2)^4$.

(c) Use appropriate technology to plot $f(x) = \ln(x)$, its tangent line, $T_1(x) = \ln(2) + \frac{1}{2}(x-2)$, and $T_4(x)$ on the same axes in the window shown Figure 8.2.6.

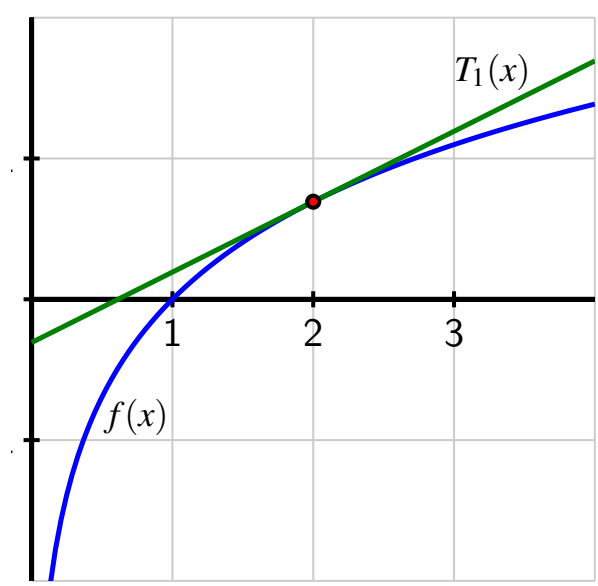


Figure 8.2.6 The function $f(x) = \ln(x)$ and its degree 1 Taylor approximation $T_1(x) = \ln(2) + \frac{1}{2}(x-2)$ near the point $(2, f(2))$.

What do you notice?

(d) Compute $|f(x) - T_4(x)|$ for several different x values (you might find it helpful to use a slider in *Desmos*); for approximately what values of x is it true that $|f(x) - T_4(x)| < 0.1$?

(e) Use the patterns you observe in parts (a) and (b) to conjecture formulas for $T_5(x)$ and $T_6(x)$.

For about what interval of x-values is it true that $|f(x) - T_5(x)| < 0.1$? What about $|f(x) - T_6(x)| < 0.1$? How is this different from what we observed with the Taylor approximations centered at $a = 1$ in Activity 8.2.3? How is it similar?

8.3 Geometric sums

Preview Activity 8.3.1. Let's explore sums of the form

$$S_n = 1 + \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 + \cdots + \left(\frac{1}{2}\right)^{n-1}.$$

Observe that we can think of the first term, 1, as $(\frac{1}{2})^0$, and since the powers of $\frac{1}{2}$ include every whole number from 0 to $n-1$, there are exactly n terms in the sum given by S_n.

(a) Note that $S_1 = 1$, $S_2 = S_1 + \frac{1}{2} = 1 + \frac{1}{2} = \frac{3}{2}$, and $S_3 = S_2 + \frac{1}{4} = \frac{3}{2} + \frac{1}{4} = \frac{7}{4}$. Using the fact that each subsequent value of S_n can be computed by adding one additional term to the preceding sum, complete Table 8.3.1 with the exact (fractional) value of each sum.

Table 8.3.1 Values of S_n for increasing values of n

$n = 1$	$S_1 = 1$
$n = 2$	$S_2 = \frac{3}{2}$
$n = 3$	$S_3 = \frac{7}{4}$
$n = 4$	$S_4 =$
$n = 5$	$S_5 =$
$n = 6$	$S_6 =$
$n = 7$	$S_7 =$

(b) What do you expect will be the exact fractional value of S_8? S_9?

(c) Conjecture a simple formula for S_n that is given by a single fraction where the numerator and/or denominator depend on n.

(d) What happens to the value of S_n as $n \to \infty$? Why?

(e) How do the results change if we instead consider

$$S_n = 1 + \frac{1}{4} + \left(\frac{1}{4}\right)^2 + \left(\frac{1}{4}\right)^3 + \cdots + \left(\frac{1}{4}\right)^{n-1} ?$$

Activity 8.3.2. Let $a = 1$ and $r = \frac{2}{5}$ be real numbers and consider the corresponding finite geometric series

$$S_n = 1 + \frac{2}{5} + \left(\frac{2}{5}\right)^2 + \cdots + \left(\frac{2}{5}\right)^{n-2} + \left(\frac{2}{5}\right)^{n-1}. \tag{8.3.1}$$

Our goal in this activity is to find a shortcut formula for S_n that can be written as a single fraction that does not involve a sum of n terms.

(a) Multiply both sides of Equation (8.3.1) by $r = \frac{2}{5}$. Write the new equation in the form

$$\frac{2}{5} \cdot S_n = \qquad\qquad . \tag{8.3.2}$$

(b) Now subtract Equation (8.3.2) from Equation (8.3.1), and explain why it follows that

$$S_n - \frac{2}{5} \cdot S_n = 1 - \left(\frac{2}{5}\right)^n. \tag{8.3.3}$$

(c) Solve Equation (8.3.3) for S_n to find a simple formula for S_n that does not involve adding n terms.

(d) How would your work above change if instead of the original geometric sum S_n, we considered the situation with $a = 7$,

$$S_n = 7 + 7 \cdot \frac{2}{5} + 7 \cdot \left(\frac{2}{5}\right)^2 + \cdots + 7 \cdot \left(\frac{2}{5}\right)^{n-1} ?$$

Activity 8.3.3. For each of the following infinite geometric series, determine the values of a and r, compute the partial sums S_5 and S_{10} exactly (writing each as a fraction), and if the infinite geometric series converges, find its value.

(a) $1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \cdots$

(b) $4 - 2 + 1 - \frac{1}{2} + \frac{1}{4} - \cdots$

(c) $2 + \frac{8}{3} + \frac{32}{9} + \frac{128}{27} + \cdots$

(d) $\sum_{k=0}^{\infty} 5 \cdot \left(\frac{3}{4}\right)^k$

(e) $\sum_{k=1}^{\infty} -2 \cdot \left(-\frac{2}{3}\right)^k$

Activity 8.3.4. Consider the function $f(x) = \frac{1}{1-x}$. Find the degree n Taylor polynomial approximation centered at $a = 0$ for f.

(a) To begin finding $T_n(x)$, do the usual work of computing the various derivatives of f and their respective values at $a = 0$; note that it's helpful to view $f(x)$ in the form $f(x) = (1-x)^{-1}$ so that we can easily compute the derivatives of f using the chain rule. For instance,

$$f'(x) = (-1)(1-x)^{-2}(-1)$$

where the first "-1" arises from the power rule, while the second "-1" results from the chain rule, since $\frac{d}{dx}[1-x] = -1$. In order to see key patterns that arise, it's helpful not to combine the products of numbers that arise in the various derivatives. Record the first five derivatives of $f(x)$ in the spaces provided below.

$$\begin{aligned} f(x) &= \frac{1}{1-x} = (1-x)^{-1} \\ f'(x) &= (-1)(1-x)^{-2}(-1) \\ f''(x) &= \\ f'''(x) &= \\ f^{(4)}(x) &= \\ f^{(5)}(x) &= \end{aligned}$$

(b) Next, evaluate the derivatives you determined in (a) at $a = 0$ and use these to find the values of the coefficients of the Taylor polynomial centered at $a = 0$. Record your work in the blank spaces provided.

$k = 0$	$f(0) = \frac{1}{1-0} = 1$	$c_0 = f(0) = 1$
$k = 1$	$f'(0) = \frac{(-1)}{(1-0)^2} \cdot (-1) = 1$	$c_1 = \frac{f'(0)}{1!} = \frac{1}{1!} = 1$
$k =$	$f''(0) =$	$c_2 = \frac{f''(0)}{2!} =$
$k =$	$f'''(0) =$	$c_3 =$
$k =$	$f^{(4)}(0) =$	$c_4 =$
$k =$	$f^{(5)}(0) =$	$c_5 =$

(c) What pattern do you observe in the value of c_k? State the degree 5 Taylor polynomial, $T_5(x)$, as well as the formula you expect for the general degree n Taylor polynomial, $T_n(x)$.

(d) By identifying the value of r, explain why the degree n Taylor polynomial $T_n(x)$ can be thought of as a finite geometric series.

(e) What is the Taylor series centered at $a = 0$ for $f(x) = \frac{1}{1-x}$? (As we will see in the next section, the "Taylor series" of a function is the infinite series that results by simply extending the Taylor polynomials indefinitely.)

8.4 Taylor series

Preview Activity 8.4.1. Let

$$T(x) = 3 - x + \frac{1}{3}x^2 - \frac{1}{9}x^3 + \cdots + 3 \cdot \left(-\frac{1}{3}\right)^n x^n + \cdots.$$

(a) Explain why $T(x)$ is a geometric series and identify the values of a and r.

(b) Recall that the sum of an infinite geometric series with first term a and constant ratio r is $S = \frac{1}{1-r}$, provided $|r| < 1$. Use this result to find a single fraction that represents the value of $T(x)$. For which values of x is this representation valid?

(c) Consider the function $f(x) = 9(3 + x)^{-1}$. Determine $f(0)$, $f'(0)$, $f''(0)$, and $f'''(0)$. What is the degree 3 Taylor polynomial centered at 0 for the function $f(x)$?

(d) How is your work in (c) connected to the original given series $T(x)$?

Activity 8.4.2. Consider the infinite Taylor series given by

$$\begin{aligned} T(x) &= \sum_{k=1}^{\infty} \frac{1}{k \cdot 2^k}(x-1)^k \\ &= \frac{1}{1 \cdot 2}(x-1) + \frac{1}{2 \cdot 4}(x-1)^2 + \frac{1}{3 \cdot 8}(x-1)^3 + \cdots + \frac{1}{n \cdot 2^n}(x-1)^n + \cdots \end{aligned}$$

(a) As described in the statement of the Ratio Test, let $r_n(x)$ be the ratio of the $(n+1)^{\text{st}}$ term of $T(x)$ to the nth term of $T(x)$. Find the simplest formula that you can for $r_n(x)$.

(b) Let $r(x) = \lim_{n\to\infty} r_n(x)$. Evaluate this limit to find the simplest formula you can for $r(x)$.

(c) For what values of x is $|r(x)| < 1$? What is the open interval of convergence for $T(x)$?

(d) Let $T_{10}(x)$ be the sum of the first 10 terms of $T(x)$, and let $f(x) = \ln(2) - \ln(3 - x)$. Plot $f(x)$ and $T_{10}(x)$ on the same coordinate axes in a window centered around $x = 1$. What do you notice? What does this suggest about the series $T(x)$?

Activity 8.4.3. Let $f(x) = e^x$. Our goal is to understand why the Taylor series for $f(x)$ converges for every real number x and see that the Taylor series converges to e^x.

(a) For $f(x) = e^x$, explain why $f^{(k)}(0) = 1$ for every natural number k.

(b) State the Taylor series, $T_f(x)$ centered at $a = 0$ for $f(x) = e^x$. Write $T_f(x)$ in both sigma notation and as an expanded sum.

(c) Let $r_n(x)$ be the ratio of the $(n+1)^{\text{st}}$ term to the nth term of $T_f(x)$. Find the simplest expression you can for $r_n(x)$.

(d) Let $r(x) = \lim_{n\to\infty} r_n(x)$. Evaluate this limit, and then apply the Ratio Test to say what you can conclude about the x-values for which $T_f(x)$ converges.

(e) Use a computational device to graph $f(x) = e^x$, $T_{10}(x)$, and $T_{20}(x)$ on the same axes. What do you observe?

8.5 Finding and using Taylor series

Preview Activity 8.5.1. Let

$$g(x) = \frac{1}{1+x^2}.$$

(a) Note that we can write $g(x) = (1+x^2)^{-1}$. Determine $g'(x)$, $g''(x)$, and $g'''(x)$.

(b) In (a), you found that

$$g'''(x) = \frac{24x - 24x^3}{(1+x^2)^4}.$$

What derivative rules would be needed to find $g^{(4)}(x)$?

(c) Find $g(0)$, $g'(0)$, $g''(0)$, and $g'''(0)$, and use the results to determine $T_3(x)$.

(d) Since the derivatives of g become so complicated, we consider a different approach to finding the Taylor series centered at $a = 0$ for $g(x) = \frac{1}{1+x^2}$. We take advantage of the fact that $f(x) = \frac{1}{1-x}$ has a Taylor series expansion we already know, and observe that f has an algebraic structure similar to g.

We introduce a new variable, u, and recall that

$$f(u) = \frac{1}{1-u} = 1 + u + u^2 + u^3 + \cdots + u^n + \cdots \tag{8.5.1}$$

for $|u| < 1$.

Let $u = -x^2$. What equation results from evaluating $f(-x^2)$ in Equation (8.5.1)?

(e) How does our work in (d) compare to the formula you found for $P_3(x)$ and suggest the Taylor series centered at $a = 0$ for $g(x) = \frac{1}{1+x^2}$? For what values of x do you expect this series will converge? Why?

Activity 8.5.2. Use known Taylor series and substitution and algebraic techniques to find a Taylor series representation for each of the following functions. In addition, state the interval of x-values for which you expect each Taylor series to converge.

(a) $g(x) = x^3 \sin(x^2)$

(b) $h(x) = e^{-x^4}$

(c) $p(x) = \dfrac{1}{1+5x}$

(d) $q(x) = x^2 \ln(1 + x^4)$

(e) $r(x) = \dfrac{e^{3x} - 1}{3x}$

Activity 8.5.3. In this activity we determine the Taylor series expansion for $\sin(x)$ in a different way and then also find the Taylor series for $\arctan(x)$.

(a) To find the Taylor series for $\sin(x)$ without using the definition, we consider the function $f(x) = \cos(x)$ and recall that $f'(x) = -\sin(x)$. Then we use the Power Series Differentiation Theorem on the known series for $f(x) = \cos(x)$ to find the Taylor series for $f'(x) = -\sin(x)$.

(i) Consider $f(x) = \cos(x)$ and recall that its Taylor series in sigma notation is

$$f(x) = \cos(x) = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!}.$$

Write out the first four nonzero terms of the Taylor series in expanded form.

(ii) Using the expanded form of the Taylor series for $f(x) = \cos(x)$, apply the Power Series Differentiation Theorem to find the Taylor series for $f'(x) = -\sin(x)$ in expanded form, being sure to simplify each constant coefficient in the expansion. Show the first three nonzero terms.

(iii) Now recall the series expansion of $\cos(x)$ in sigma notation. What is the derivative of the general term of the series,

$$\frac{d}{dx}\left[(-1)^k \frac{x^{2k}}{(2k)!}\right],$$

in its most simplified form?

(iv) Finally, state the Taylor series for $f'(x) = -\sin(x)$ in sigma notation. Be sure to decide whether the series should be indexed starting with $k = 0$ or $k = 1$ and to justify your choice.

(v) How can you now use your work in part (iv) to find the Taylor series for $\sin(x)$?

(b) Consider the function $g(x) = \arctan(x)$.

(i) What is $g'(x)$?

(ii) Recall that in Preview Activity 8.5.1, we found the Taylor series expansion for

$$f(x) = \frac{1}{1+x^2}.$$

State the Taylor series for $f(x)$ that you found in the preview activity.

(iii) Explain why we can think of $g(x)$ as being given by

$$g(x) = \int_0^x f(u)\,du.$$

Use this relationship to find a power series expansion for $g(x)$.

(iv) For what interval of x-values is the Taylor series for $g(x) = \arctan(x)$ guaranteed to converge?

Activity 8.5.4. In this activity we will determine the Taylor series expansion for the famous *error function*,

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2}\, dt,$$

that arises in probability theory and other important applications.

(a) Use the Taylor series for e^x to find the Taylor series for e^{-t^2}.

(b) Next, evaluate the integral $\int_0^x e^{-t^2}\, dt$ by replacing e^{-t^2} with its Taylor series.

(c) Use your work in (b) to state the Taylor series for $\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2}\, dt$

(d) For what interval of x-values will the Taylor series for $\operatorname{erf}(x)$ converge? Why?

(e) In probability theory, $\operatorname{erf}(x)$ is important because of its connection to the *normal distribution*, which is represented by a bell curve. Indeed, $\operatorname{erf}(x)$ represents the fraction of a normally distributed characteristic in a population that lies between 0 and x. How can you use your result in (c) to estimate $\operatorname{erf}(0.5)$?

8.6 Quantifying the accuracy of approximations

Preview Activity 8.6.1. Consider the alternating geometric series

$$S = \sum_{k=0}^{\infty} (-1)^k \left(\frac{4}{5}\right)^k = 1 - \frac{4}{5} + \frac{16}{25} - \cdots + (-1)^{n-1}\left(\frac{4}{5}\right)^{n-1} + \cdots.$$

We want to explore how the partial sums of the series compare to and approximate the exact sum of the series, which is

$$S = \frac{a}{1-r} = \frac{1}{1-\left(\frac{-4}{5}\right)} = \frac{1}{\frac{9}{5}} = \frac{5}{9}.$$

(a) Recall that the nth partial sum, S_n, is the sum of the first n terms of the infinite geometric series S. This means that

$$S_n = \sum_{k=0}^{n-1} (-1)^k \left(\frac{4}{5}\right)^k = 1 - \frac{4}{5} + \frac{16}{25} - \cdots + (-1)^{n-1}\left(\frac{4}{5}\right)^{n-1}.$$

Note that the exact fractional values of S_1, S_2, ..., S_6, have been recorded below along with their decimal representations. Your task is to use this information to do some related computations that help us understand the behavior of the series and its partial sums.

First, by computing the differences between S_n and S for several different values of n (recalling that $S = \frac{5}{9} = 0.\overline{5}$), fill in the first column of blank spaces provided below. Then, by viewing the series $S = 1 - \frac{4}{5} + \frac{16}{25} - \cdots + (-1)^{n-1}\left(\frac{4}{5}\right)^{n-1} + \cdots$ as being in the form $S = a_0 + a_1 + a_2 + \cdots$, fill in the last column of blank spaces.

$n = 1$	$S_1 = 1$	$S_1 - S =$	$a_1 = -0.8$
$n = 2$	$S_2 = \frac{1}{5} = 0.2$	$S_2 - S =$	$a_2 = 0.64$
$n = 3$	$S_3 = \frac{21}{25} = 0.84$	$S_3 - S =$	$a_3 =$
$n = 4$	$S_4 = \frac{41}{125} = 0.328$	$S_4 - S =$	$a_4 =$
$n = 5$	$S_5 = \frac{461}{625} = 0.7376$	$S_5 - S =$	$a_5 =$
$n = 6$	$S_6 = \frac{1281}{3125} = 0.40992$	$S_6 - S =$	$a_6 =$

(b) What do you notice about the differences between S_n and S as the value of n increases? There are at least two important things you can say.

(c) What do you notice about how the differences between S_n and S compare to the value of a_n? There are at least two important things you can say.

Activity 8.6.2. In this activity we approximate the values of several different alternating series using the Alternating Series Estimation Theorem.

(a) Use the fact that $\sin(x) = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \cdots$ to estimate $\sin(1)$ to within 0.0001. Do so without entering "sin(1)" on a computational device. After you find your estimate, enter "sin(1)" on a computational device and compare the results.

(b) Recall our recent work with $\int_0^1 e^{-x^2}\,dx$ which can be expressed as the series

$$\int_0^1 e^{-x^2}\,dx = 1 - \frac{1}{3} + \frac{1}{2!\cdot 5} - \frac{1}{3!\cdot 7} + \cdots + \frac{(-1)^n}{n!\cdot(2n+1)} + \cdots.$$

Use this series representation to estimate $\int_0^1 e^{-x^2}\,dx$ to within 0.0001. Then, compare what a computational device reports when you use it to estimate the definite integral.

(c) Find the Taylor series for $\cos(x^2)$ and then use the Taylor series and to estimate the value of $\int_0^1 \cos(x^2)\,dx$ to within 0.0001. Compare your result to what a computational device reports when you use it to estimate the definite integral.

(d) Recall we know that if $|x| < 1$, then

$$\ln(1+x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \cdots + (-1)^{n-1}\frac{1}{n}x^n + \cdots.$$

What happens if $x = 1$?

Explain why the series $1 - \frac{1}{2}\cdot 1^2 + \frac{1}{3}\cdot 1^3 - \cdots + (-1)^{n-1}\frac{1}{n}\cdot 1^n + \cdots$ must converge and estimate its sum to within 0.01. What is the exact sum of this series?

Activity 8.6.3. In this activity we apply the Lagrange Error Bound to quantify the accuracy of several different approximations.

(a) Use the degree 10 Taylor polynomial (centered at $a = 0$) of $f(x) = e^x$ to estimate the value of e^2. What is the maximum error of your estimate, according to the Lagrange Error Bound? How does this compare to the actual error between e^2 and $T_{10}(2)$ as reported by a computer algebra system?

(b) Use a degree n Taylor polynomial (centered at $a = 0$) of $f(x) = \cos(x)$ to estimate the value of $\cos(1)$ within an accuracy of 0.00000001. According to the Lagrange Error Bound, what value of n is needed to achieve this accuracy? What is the resulting approximate value of $\cos(1)$?

(c) Recall that for $f(x) = \ln(1+x)$, its Taylor series centered at $a = 0$ is given by

$$\ln(1+x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \cdots + (-1)^{n-1}\frac{1}{n}x^n + \cdots$$

and that the nth derivative of $f(x) = \ln(1+x)$ is given by

$$f^{(n)}(x) = \frac{(-1)^{n-1}(n-1)!}{(1+x)^n}.$$

If we want to estimate $f(0.5) = \ln(1.5)$ to within an accuracy of 0.0001, what value of n is needed to achieve this accuracy from computing $T_n(0.5)$, according to the Lagrange Error Bound?

APPENDIX A

A Short Table of Integrals

a. $\displaystyle\int \frac{du}{a^2+u^2} = \frac{1}{a}\arctan\left(\frac{u}{a}\right) + C$

b. $\displaystyle\int \frac{du}{\sqrt{u^2 \pm a^2}} = \ln|u + \sqrt{u^2 \pm a^2}| + C$

c. $\displaystyle\int \sqrt{u^2 \pm a^2}\,du = \frac{u}{2}\sqrt{u^2 \pm a^2} \pm \frac{a^2}{2}\ln|u + \sqrt{u^2 \pm a^2}| + C$

d. $\displaystyle\int \frac{u^2 du}{\sqrt{u^2 \pm a^2}} = \frac{u}{2}\sqrt{u^2 \pm a^2} \mp \frac{a^2}{2}\ln|u + \sqrt{u^2 \pm a^2}| + C$

e. $\displaystyle\int \frac{du}{u\sqrt{u^2+a^2}} = -\frac{1}{a}\ln\left|\frac{a+\sqrt{u^2+a^2}}{u}\right| + C$

f. $\displaystyle\int \frac{du}{u\sqrt{u^2-a^2}} = \frac{1}{a}\operatorname{arcsec}\left(\frac{u}{a}\right) + C$

g. $\displaystyle\int \frac{du}{\sqrt{a^2-u^2}} = \arcsin\left(\frac{u}{a}\right) + C$

h. $\displaystyle\int \sqrt{a^2-u^2}\,du = \frac{u}{2}\sqrt{a^2-u^2} + \frac{a^2}{2}\arcsin\left(\frac{u}{a}\right) + C$

i. $\displaystyle\int \frac{u^2}{\sqrt{a^2-u^2}}\,du = -\frac{u}{2}\sqrt{a^2-u^2} + \frac{a^2}{2}\arcsin\left(\frac{u}{a}\right) + C$

j. $\displaystyle\int \frac{du}{u\sqrt{a^2-u^2}} = -\frac{1}{a}\ln\left|\frac{a+\sqrt{a^2-u^2}}{u}\right| + C$

k. $\displaystyle\int \frac{du}{u^2\sqrt{a^2-u^2}} = -\frac{\sqrt{a^2-u^2}}{a^2u} + C$

Colophon

This book was authored in PreTeXt.